

Greedy Algorithm

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• Introduction of Greedy algorithm —

→ Optimization:

Given a problem if I try to minimize or maximize there property then it is called optimization problem.

Optimization Problems are

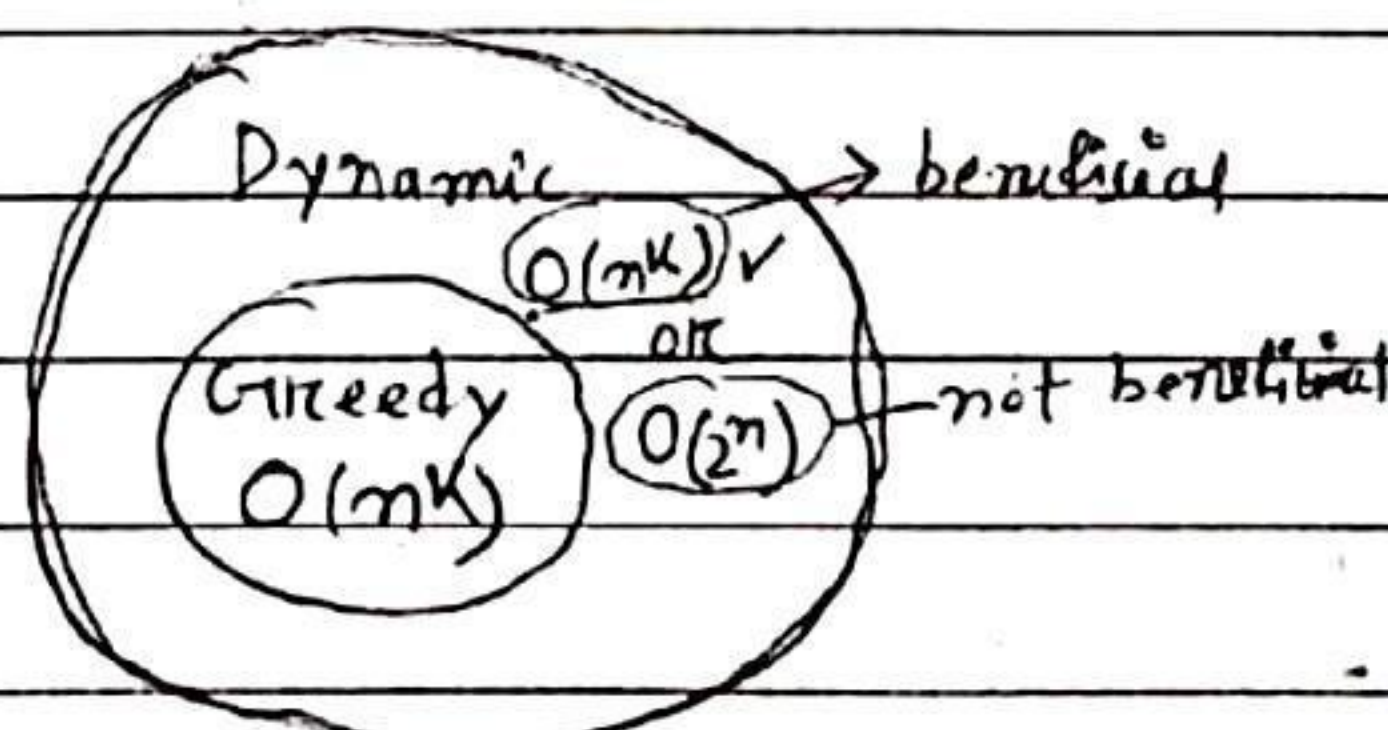
- minimize cost.
- max profit.
- maximize reliability.
- minimize risk.

→ Greedy method and dynamic programming are two programming paradigms which can be used to solve optimization problem.

→ Using Greedy method we will not be able to solve all the optimization problem.

→ Dynamic Dynamic program can solve any optimization problem. (we want to apply dynamic programming if can do something better than compare to exhaustive search)

→ Time complexity of dynamic programming could be — $O(n^k)$ or $O(2^n)$

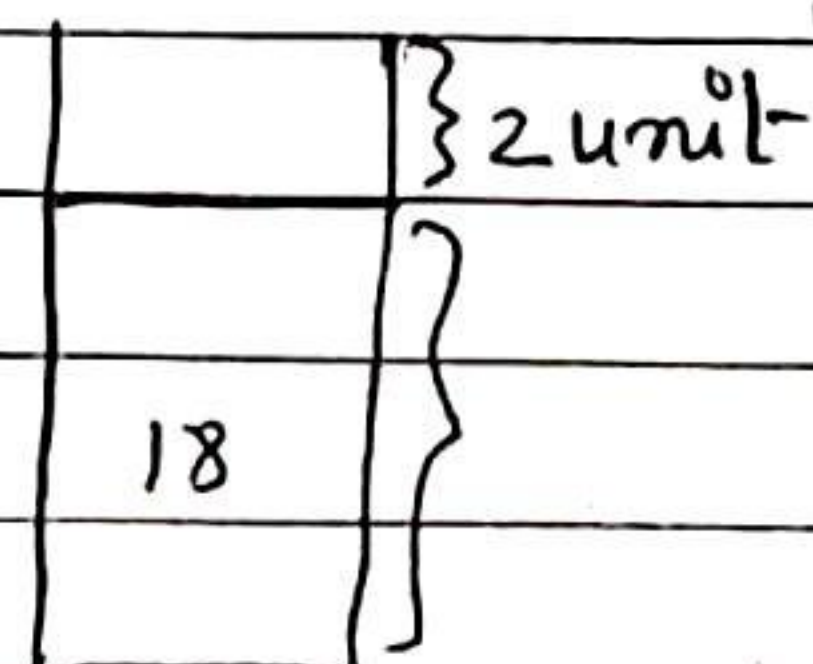


- ~~Greedy Knapsack algorithm~~
- Knapsack problem:
→ means bag

	Obj-1	Obj-2	Obj-3	$M=20$ (M - capacity of bag)
profit	25	24	15	$n=3$ (n - no. of object)
weight	18	15	10	

When
→ Greedy about profit -

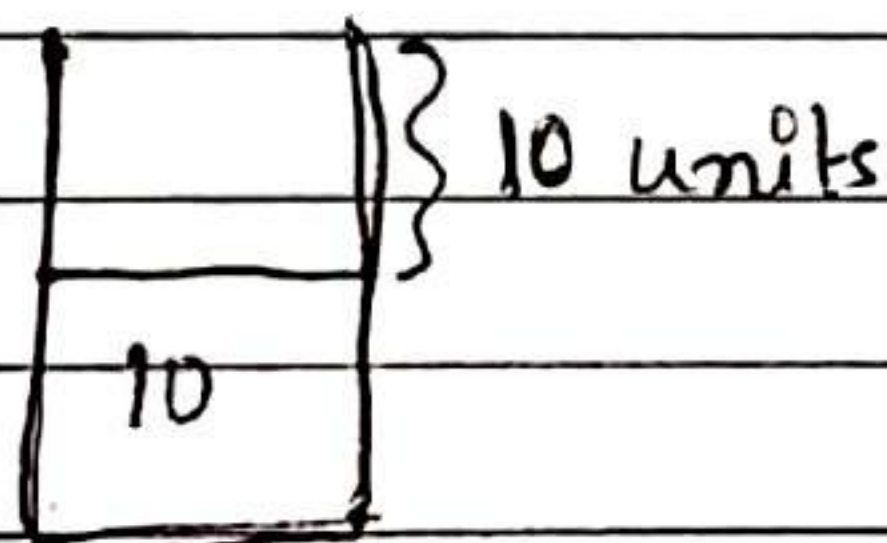
$$\frac{15W - 24P}{2W = 24 \times \frac{2}{15}}$$



20 units

	Weight	Profit
Obj-1:	18	25
Obj-2:	2	$(24) \times (\frac{2}{15})$
	20	<u>28.2</u>

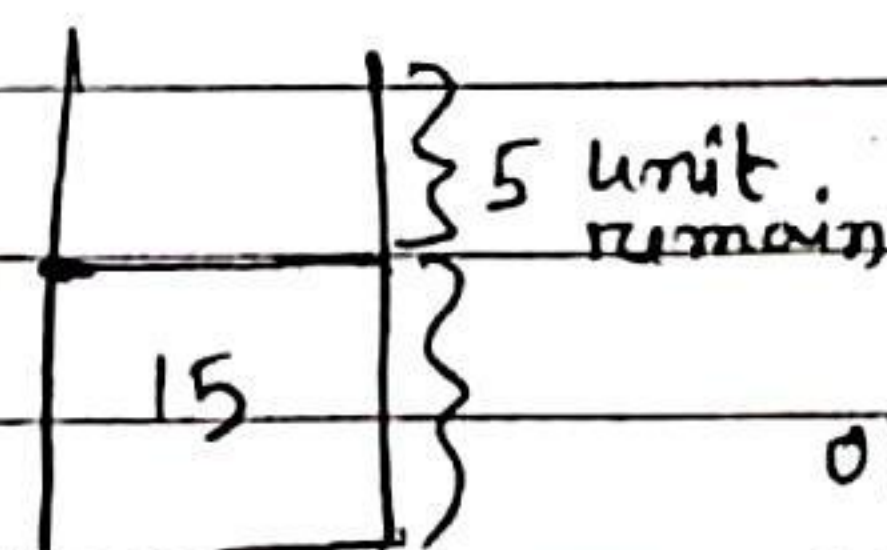
→ When Greedy about weight -



	W	P
Obj-3:	10	15
Obj-2:	10	$(24) \times (\frac{10}{15})$
	20	<u>31</u>

→ When Greedy about ratio of profit & weight.

profit per unit, Obj-1: $\frac{25}{18} = 1.4$
Obj-2: $\frac{24}{15} = 1.6$



Obj-3: $\frac{15}{10} = 1.5$

	W	Profit
Obj-2:	15	24
Obj-3:	5	$(15) \times (\frac{5}{10})$
	20	<u>31.5</u>

ratio of profit

When Greedy about Λ
more Profit Λ
compare to

Algorithm =

Greedy Knapsack

{

for $i=1$ to n ;- Compute P_i/w_i ; — $O(n)$ - sort objects in non increasing order of P/w .for $i=1$ to n from sorted list:if $(m > 0 \ \&\& \ w_i \leq M)$ • $M = M - w_i$; $P = P + P_i$;} $O(n)$

else

break;

if $(m > 0)$ $P = P + P_i \left(\frac{M}{w_i} \right)$;

{

sorting algo take
(merge sort) or
 $O(n \log n)$ heap sortEx: 1 What is max profit get out of it - $M=15, n=7$

	$i=1$	2	3	4	5	6	7
Objects	1	2	3	4	5	6	7
profits	10	5	15	7	6	18	3
Weight	2	3	5	7	①	4	1
(1) $\frac{P_i}{w_i} =$	5	1.6	3	1	6	4.5	3

(11)

5	1	6	3	7	2	4
---	---	---	---	---	---	---

Profit	2
	7
	3
	6
	1
	5

 $M=15$ units. $M = 15 \ 14 \ 12 \ 8 \ 3 \ ② \ 0$

$$P = 6 + 10 + 18 + 15 + 3 + 5 \left(\frac{2}{3} \right)$$

$$= 55.3$$

Time Complexity, $= O(n \log n)$.

$$= O(n) + O(n \log n) + O(n) + O(1)$$

$$= \boxed{O(n \log n)}$$

Ex: 2 What is max profit get out of it -
 $M = 15, n = 5$

Objut -	1	2	3	4	5
P -	2	28	25	18	9
W -	1	4	5	3	3
P/W -	2	7	5	6	3

Sort them -

- In decreasing -

- Order

Objut	Ob	Ob	Ob	Ob
2	4	3	5	1

	S
Ob -	3
Ob -	4
Ob -	2

$$M = 15 \quad 18 \quad 8 \quad 3 \quad 0$$

$$P = 28 + 18 + 25 + 9$$

$$= \boxed{80}$$

Huffman Coding:

→ Huffman coding is an algorithm which is used to find the codes using which you could actually compress the size of data.

		uniform encoding	frequency	uniform encoding
100 character file	a b c d	a - 00 b - 01 c - 10 d - 11	a - 50 b - 40 c - 5 d - 5	a - 0 b - 10 c - 110 d - 111

$$(100) \times 2 = 200 \text{ bits required to 100 character file.}$$

a b c d a
0 10 110 111 0
a b c d a

$$(50) \times 1 + (40) \times 2 + 5 \times (3) + 5 \times (3) = 180 \text{ bits required for 100 character file.}$$

HUFFMAN (C)

{

$$n = |C|$$

Make a min heap Q with C — $O(n)$

for $i = 1$ to $n-1$

(allocate a new node z)

$$z.\text{left} = x = \text{Extract-min}(Q) \quad \left. \begin{array}{l} z.\text{right} = y = \text{Extract-min}(Q) \\ z.\text{freq} = x.\text{freq} + y.\text{freq} \end{array} \right\} \begin{array}{l} (n-1) \times 2 \times \log n \\ = (n \log n) \end{array}$$

$$z.\text{right} = y = \text{Extract-min}(Q)$$

$$z.\text{freq} = x.\text{freq} + y.\text{freq}$$

$$\text{Insert}(Q, z) \quad \begin{array}{l} (n-1) \times \log n \\ = (n \log n) \end{array}$$

$$\text{return}(\text{Extract-min}(Q))$$

}

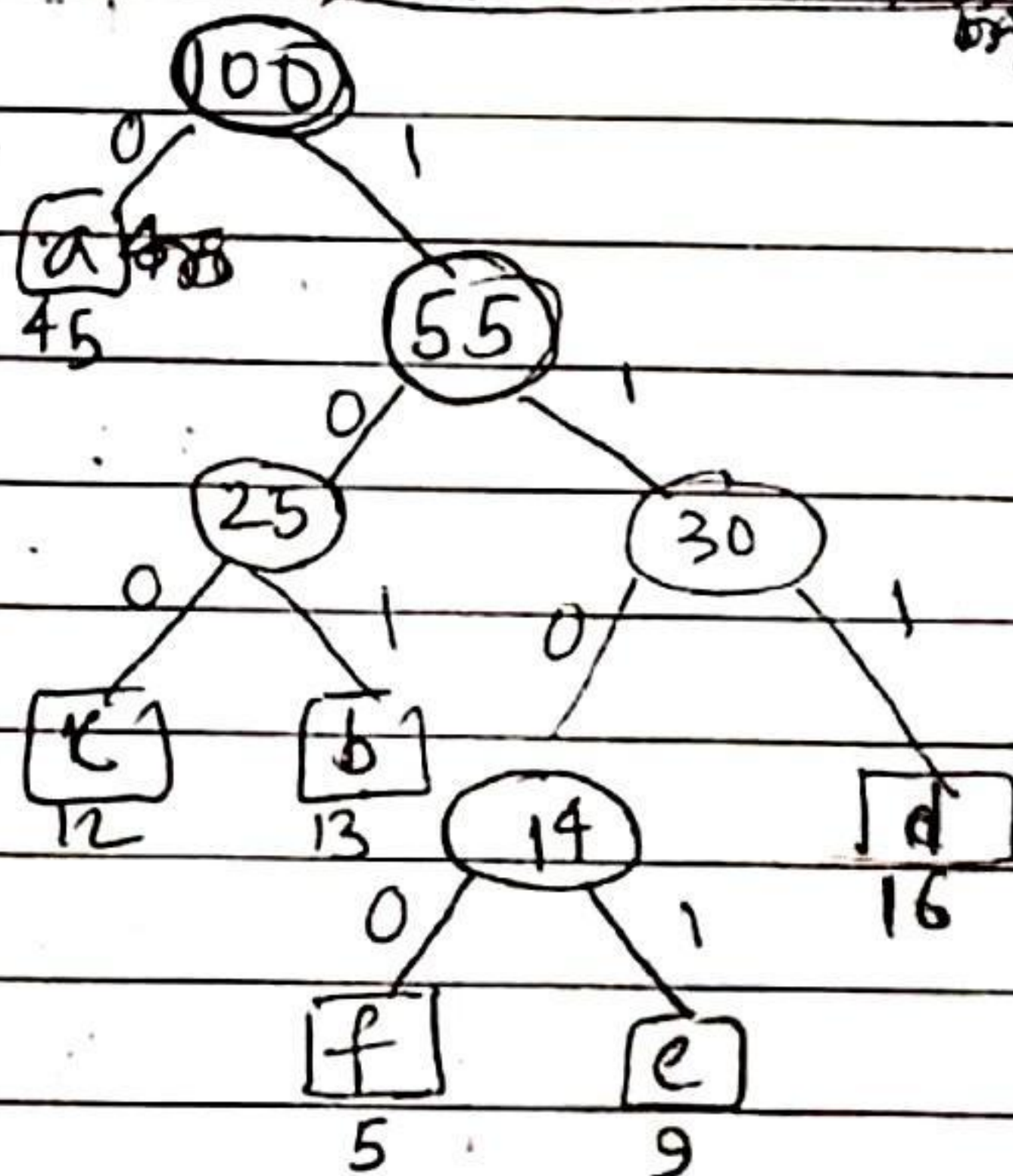
$$O(1)$$

Ex: 1

no. of character, $n = 6$

char -	a	b	c	d	e	f
freq -	45	13	12	16	9	5

Q: [45, 13, 12, 16, 9, 5]



45	13	12	16	14
25		30		

all char got diff uniformed

a = 0, code.

b = 101, 3bit

c = 100,

d = 111,

e = 1101, 4bit

f = 1100,

$$a - 45 \times 1\text{bit} = 45$$

$$b - 13 \times 3\text{bit} = 39$$

$$c - 12 \times 3\text{bit} = 36$$

$$d - 16 \times 3 = 48$$

$$e - 9 \times 4\text{bit} = 36$$

$$f - 5 \times 4 = 20$$

224 bits required
to save file.

→ Time complexity = $O(n \log n)$.

→ Space complexity = $O(n)$.

{for delete and insert heap better than sort
for only delete sorting algo is better.

Ex: 2

Suppose the letters a, b, c, d, e, f have probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{32}$.

1) Which of the following is the Huffman code for the letters a, b, c, d, e, f?

☒ a) 0, 10, 110, 1110, 11110, 11111.

☐ b) 11, 10, 011, 010, 001, 000.

☐ c) 11, 10, 01, 001, 0001, 0000.

☐ d) 110, 100, 010, 000, 001, 111.

2) What is the average length of the correct answer to question 1?

☐ a) 3

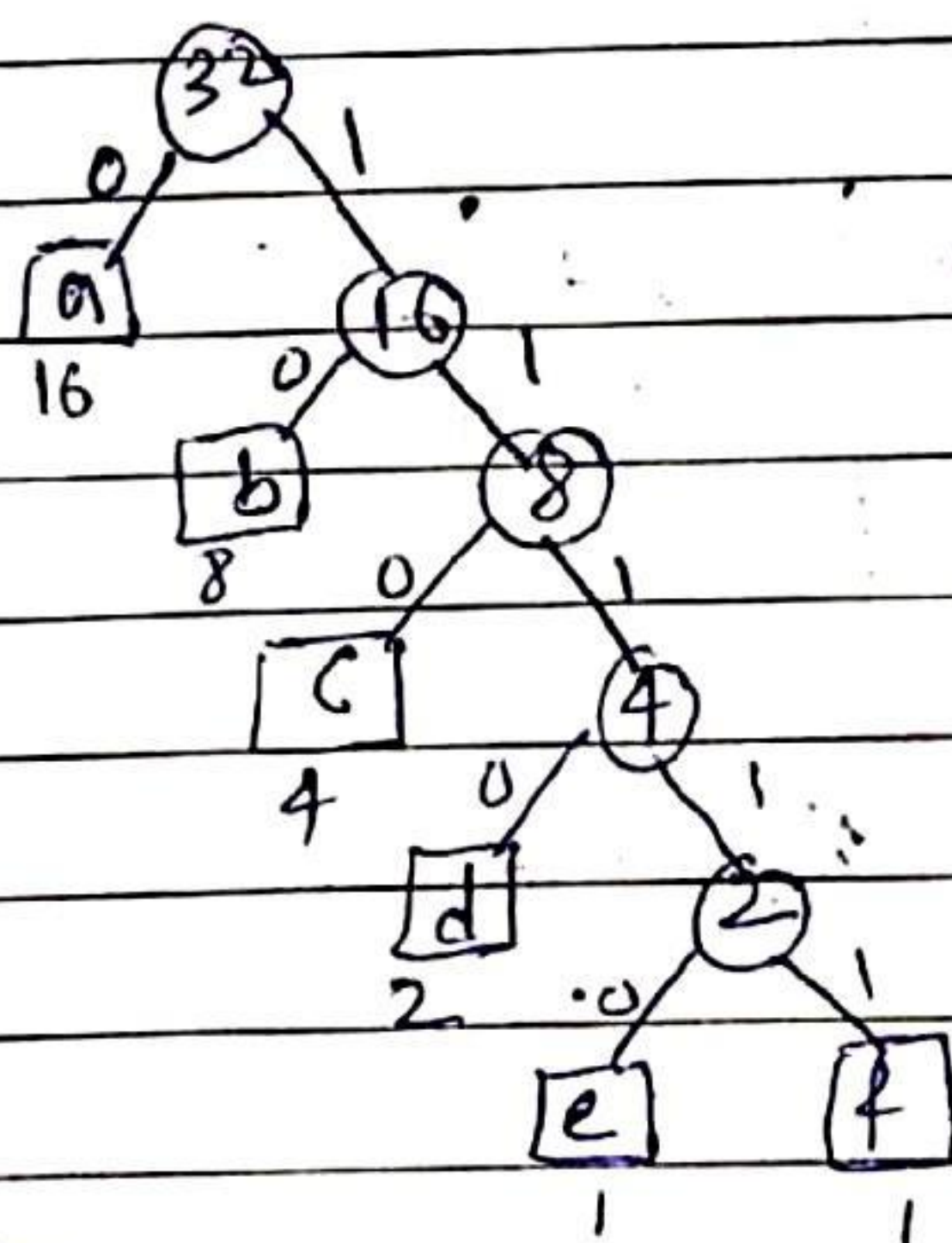
☐ b) 2.1875

☐ c) 2.25

☒ d) 1.9375.

→ (4-1)

	a	b	c	d	e	f
mul by 32	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{32}$
	16	8	4	2	1	1



a - 0

b - 10

c - 110

d - 1110

e - 11110

f - 11111

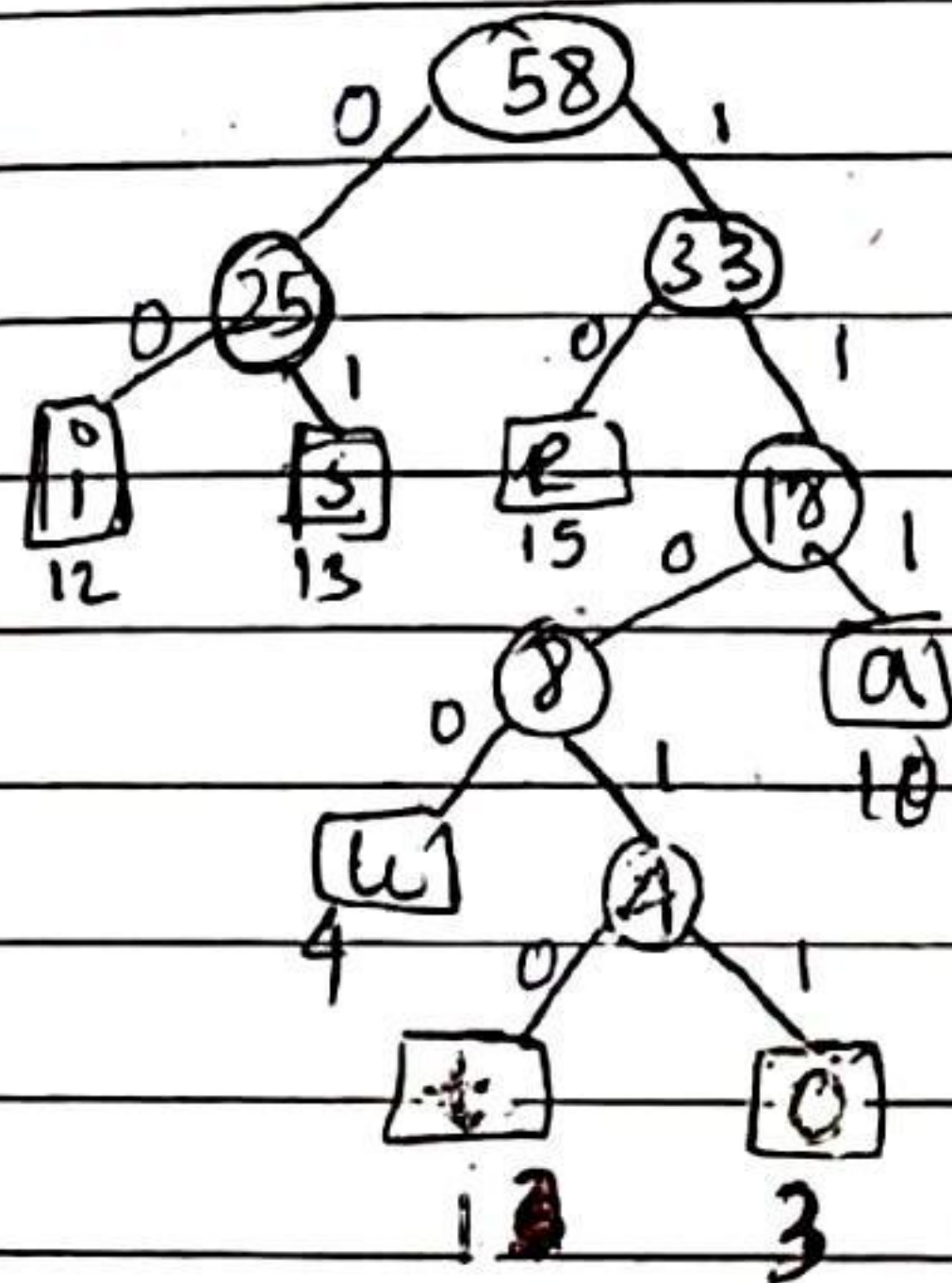
Small - left
large - Right

(4-2)

$$\frac{16 \times 1 + 8 \times 2 + 4 \times 3 + 2 \times 4 + 1 \times 5 + 1 \times 5}{16 + 8 + 4 + 2 + 1 + 1} = 1.9375$$

ex: 3

	a	e	i	o	u	s	t
freq of char	10	15	12	3	4	13	1



10 15 12 ~~3~~ 4 13 ~~1~~
 10 15 12 ~~3~~ 13 ~~1~~
~~10~~ 15 12 13 ~~1~~

a - 111

e - 10

i - 00

o - 11011

u - 1100

s - 01

t - 11010

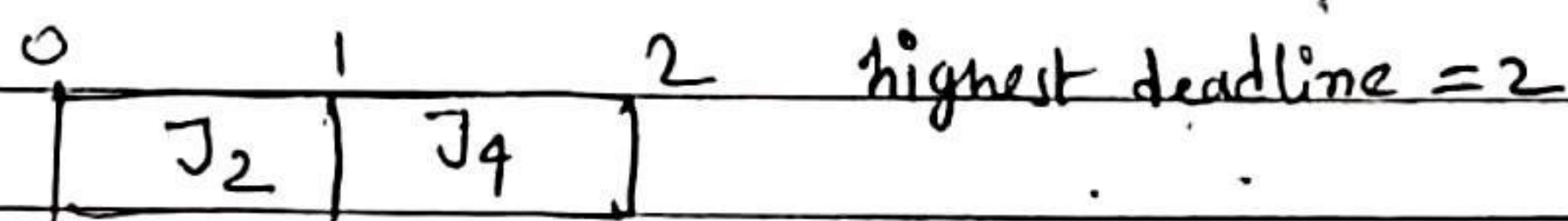
18 15 ~~12~~ 13~~18~~ 15 25

33 25

• Job Sequencing with Deadlines =

ex: 1

	JOB-1	J-2	J-3	J-4
Deadline:	2	1	1	2
profit :	6	(8)	5	(10)

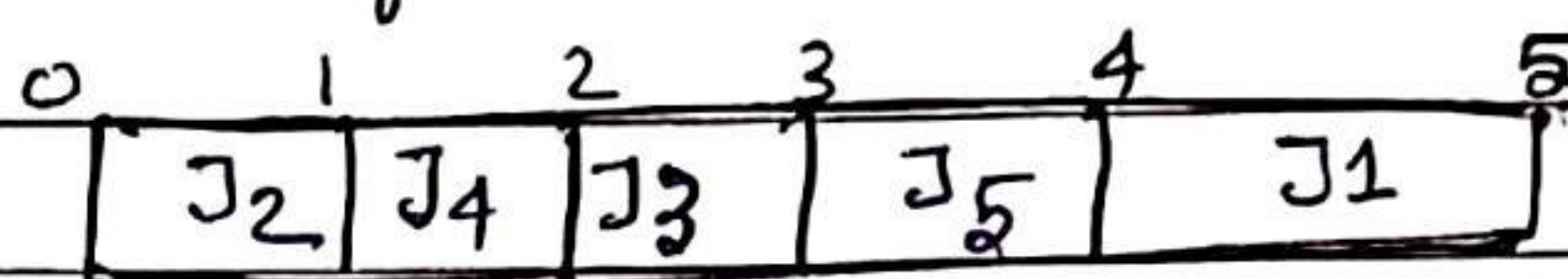


profit = 18

ex: 2

	JOB-1	J2	J3	J4	J5	J6
Deadline:	5	3	3	2	4	2
profit:	200	180	190	(300)	120	100

high Max deadline = 5

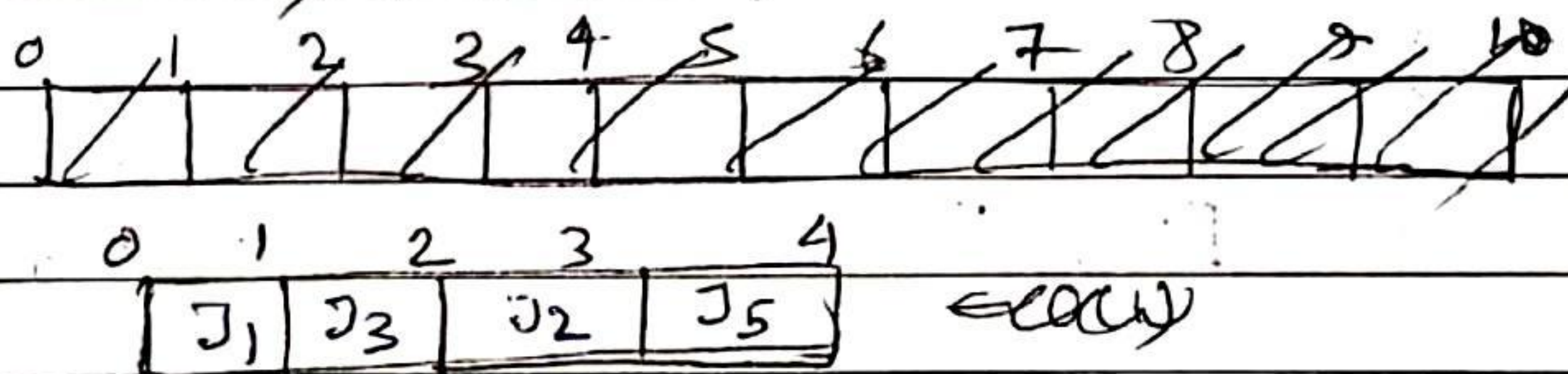


Time complexity to such algo = $O(n \times n)$
 $= n^2$

ex: 3

Jobs:	J1	J2	J3	J4	J5
profits:	(2)	(4)	(3)	1	(10)
deadlines:	3	3	3	4	4

Max deadline = 4

profit = ~~10~~ 2 + 4 + 3 + 10 = 19Time complexity: $O(n^2)$ creation of array - $O(1)$ sort the job by profit in descending order
+ $- O(n \log n)$ To put n job in array and travel
left $(n) - O(n \times n) = O(n^2)$

Ex: 4

Jobs	J ₁	J ₂	J ₃	J ₄	J ₅	J ₆	J ₇	J ₈	J ₉
profits	(15)	(20)	(30)	(18)	(18)	(10)	(23)	(16)	(25)
deadlines	7	2	5	3	4	5	2	7	3

Max deadline time = 7

0	1	2	3	4	5	6	7
J ₂	J ₇	J ₉	J ₅	J ₃	J ₁	J ₈	

$\checkmark$ J₃, $\checkmark$ J₉, $\checkmark$ J₇, $\checkmark$ J₂, $\times$ J₄, $\checkmark$ J₅, $\checkmark$ J₈, $\checkmark$ J₁, $\times$ J₆

$$\text{profit} = 30 + 25 + 23 + 20 + 18 + 16 + 15 = 147$$

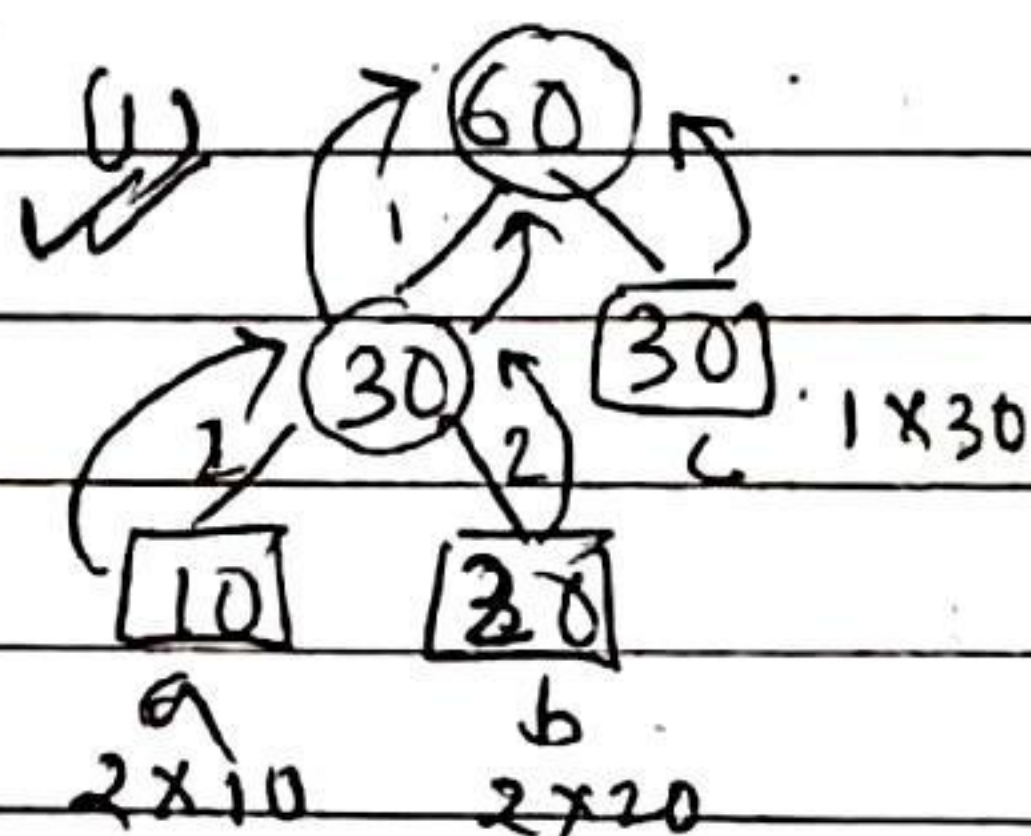
Time complexity = $O(n^2)$

Optimal Merge patterns =

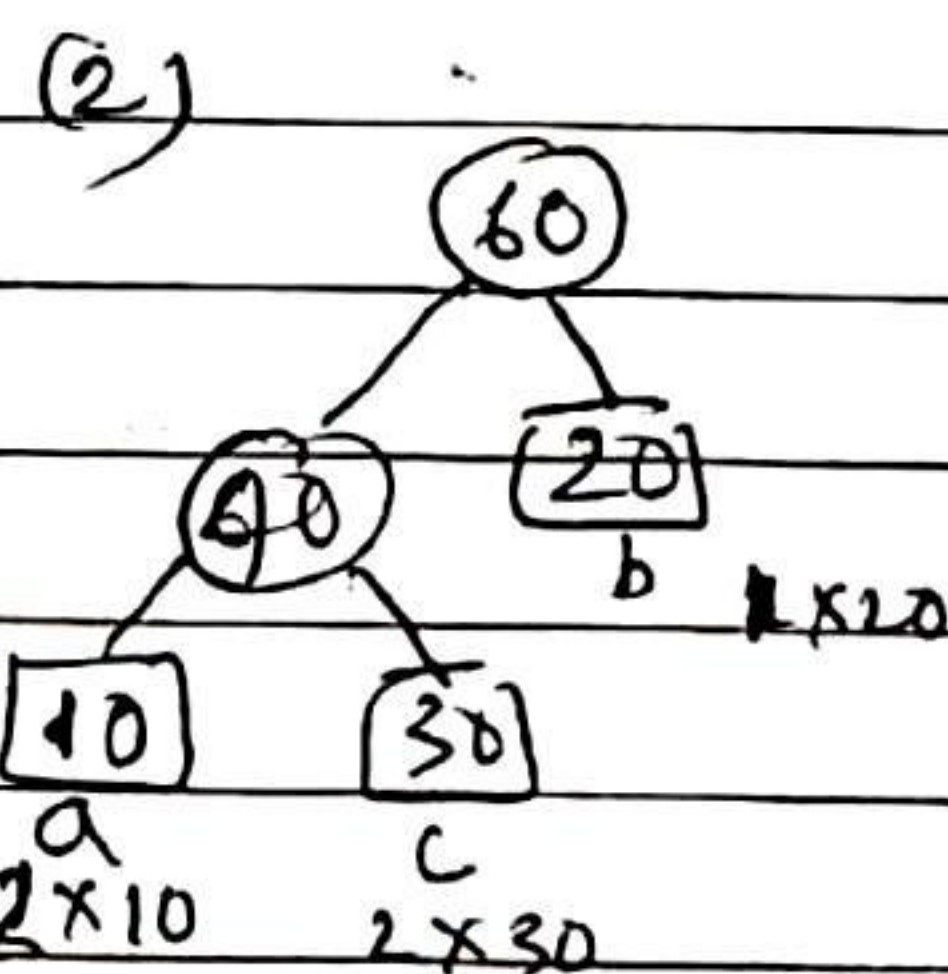
Ex: 1

files	a	b	c
records	[10]	[20]	[30]

• find minimum record movements?



$$\text{Total record movement} = 20 + 40 + 30 = 90$$

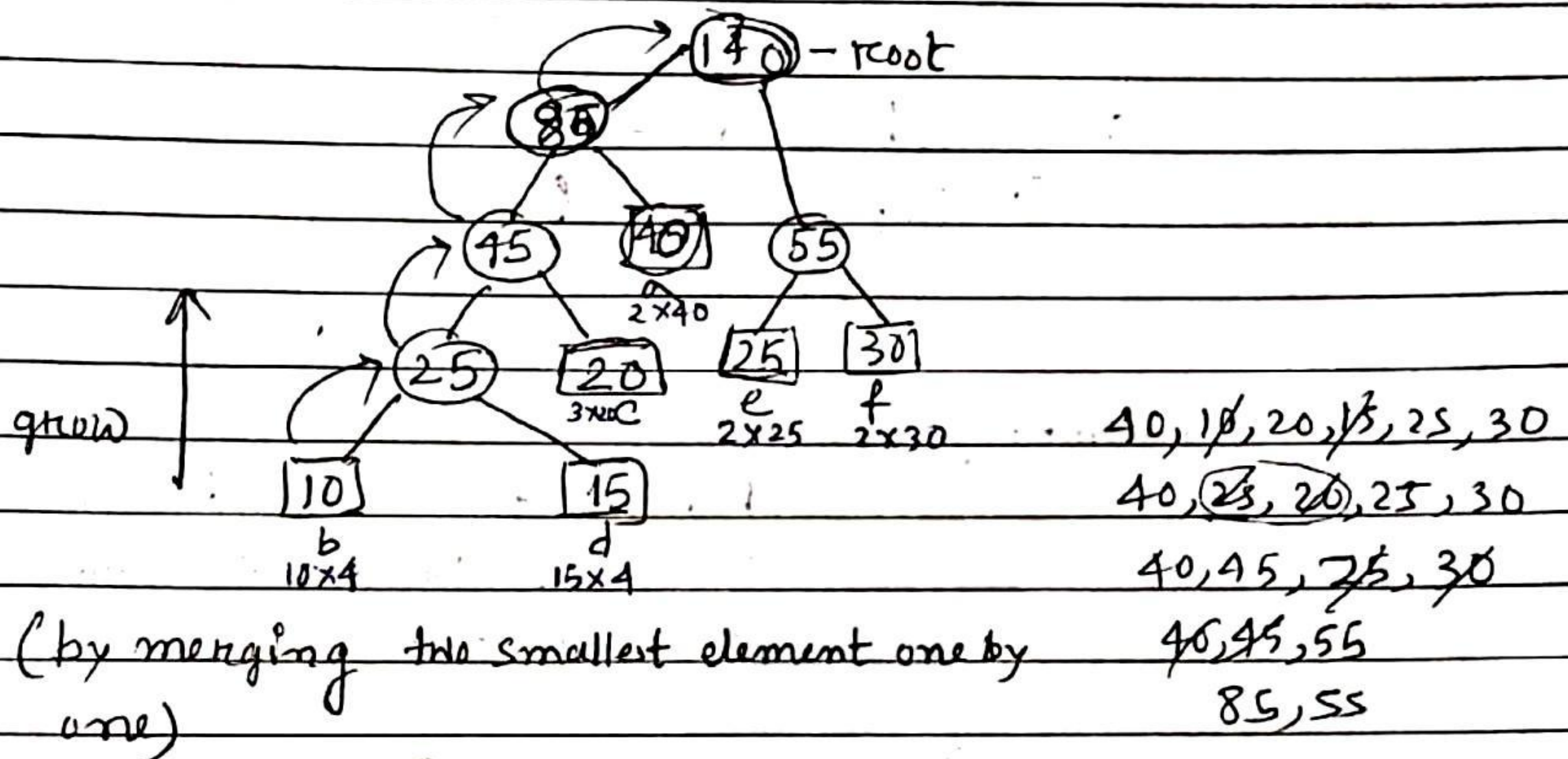


$$\text{Total record movement} = 20 + 60 + 20 = 100$$

- By merging least two nodes. (Huffman coding)

ex: 2)

files -	a	b	c	d	e	f
records -	40	10	20	15	25	30



Total record movement required,

$$\begin{aligned}
 &= 10 \times 4 + 15 \times 4 + 3 \times 20 + 2 \times 40 + 2 \times 25 + 2 \times 30 \\
 &= 40 + 60 + 60 + 80 + 50 + 60 \\
 &= \boxed{350}
 \end{aligned}$$

Time Complexity = $O(n \log n)$ (like Huffman coding algorithm)

• Introduction, Spanning trees and Kirchhoff theorem:

• GRAPH: (V, E)

→ collection of vertices and edges call graph.

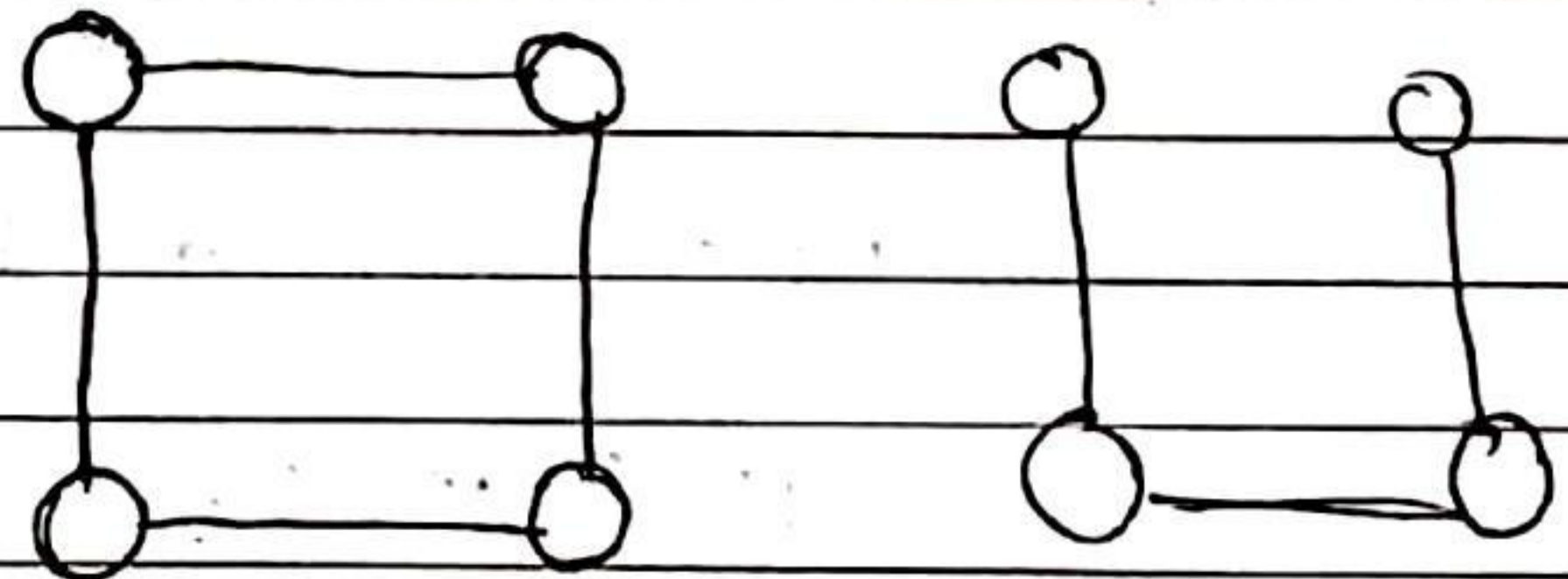
Graph two type -

① Simple graph.

② Multiple graph.

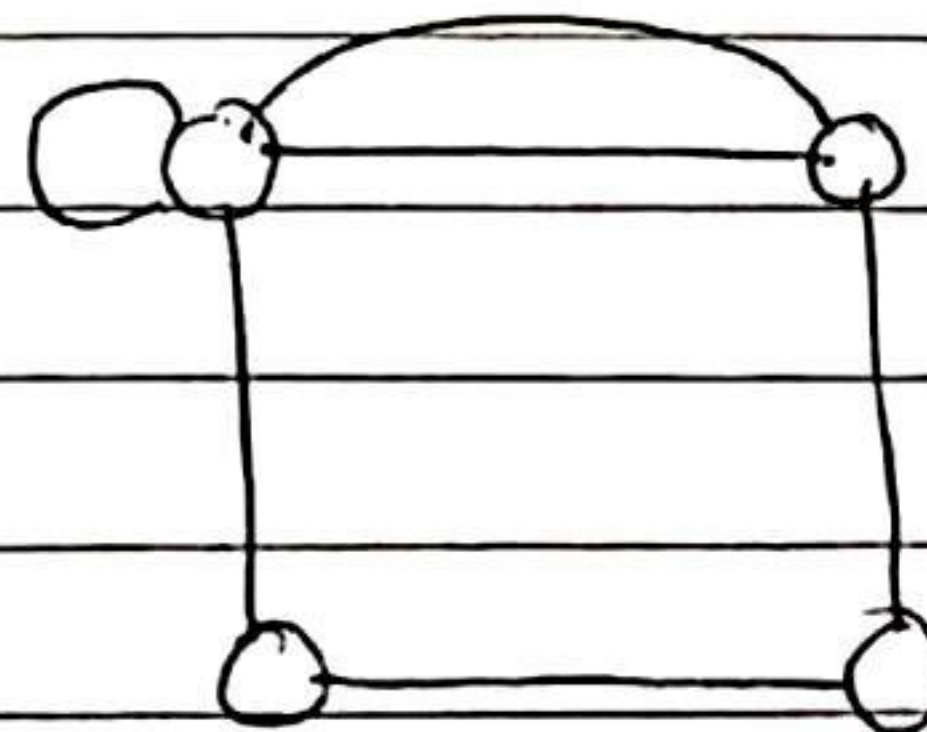
① Simple graph -

between two vertices, if there is at most one edge present then that is called simple graph.

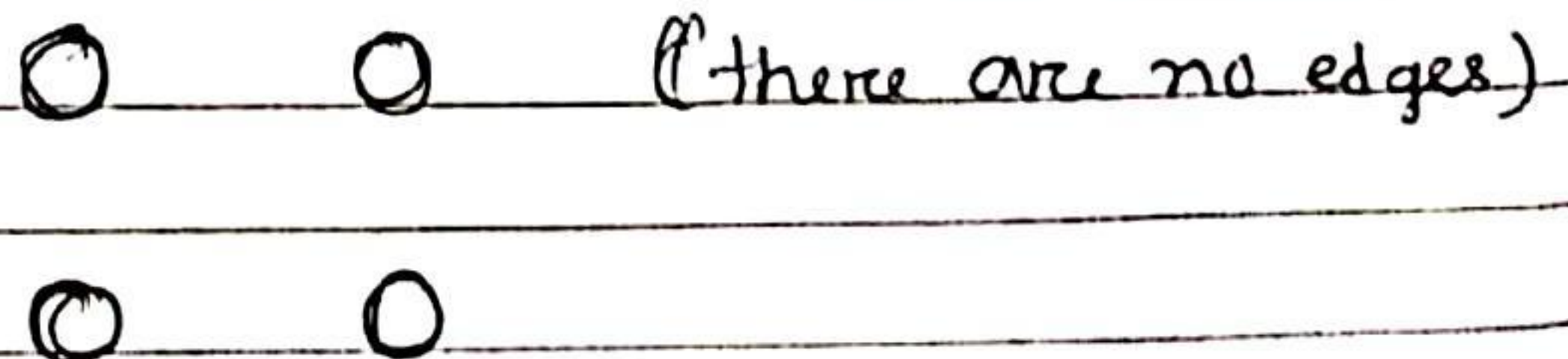


② Multi graph -

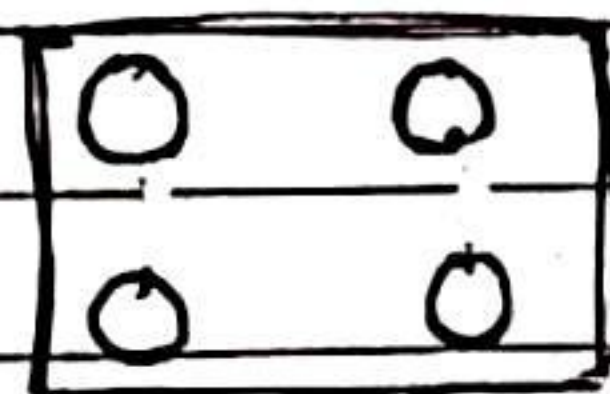
between two vertices, if there is more than one edge present then that is called multi graph.



NULL graph -

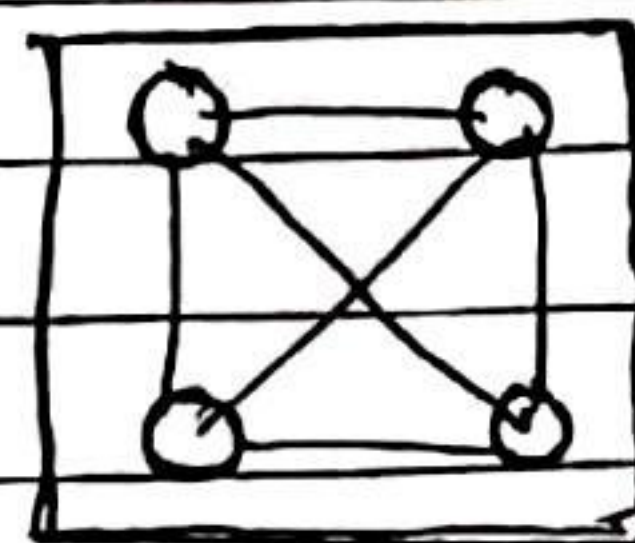


→ minimum no. of edges present in a simple graph is = 0



$$\min(V) = 0$$

→ Max no. of edges present in a simple graph is =



every 2 node
there are 1 edges

$$\boxed{\max = \frac{n(n-1)}{2}}$$

($n \rightarrow$ no. of vertices)

($E \rightarrow$ Edge)

$$= \frac{n(n-1)}{2}$$

$$= \frac{n(n-1)}{2}$$

$$= \frac{n^2 - n}{2}$$

$$\left\{ \begin{array}{l} \text{In worst no. of edge } E = O(n^2) \\ \text{" " " vertices } V = O(\log E) \end{array} \right\} \quad \begin{array}{l} E = O(n^2) \\ = O(V^2) \end{array}$$

$$\log E = \log V^2$$

$$V = (\log_2 E)$$

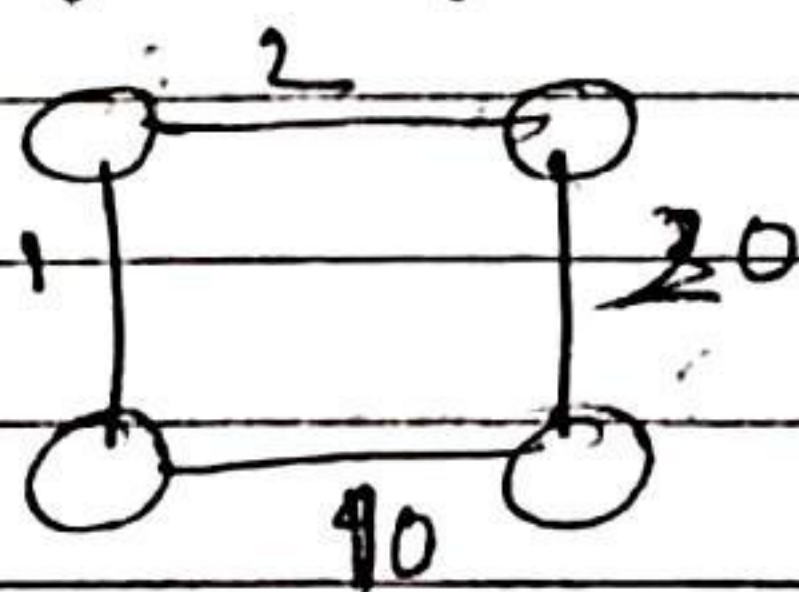
• Spanning trees = (min E required to connect n vertices)

Other classification of graph

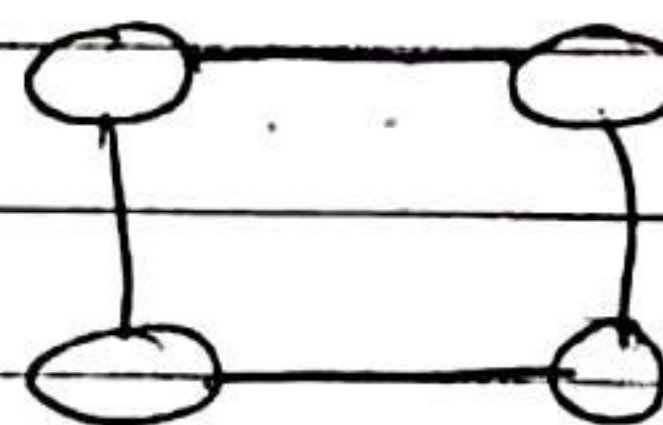
(i) Weighted graph.

(ii) Unweighted graph.

① Weighted graph —

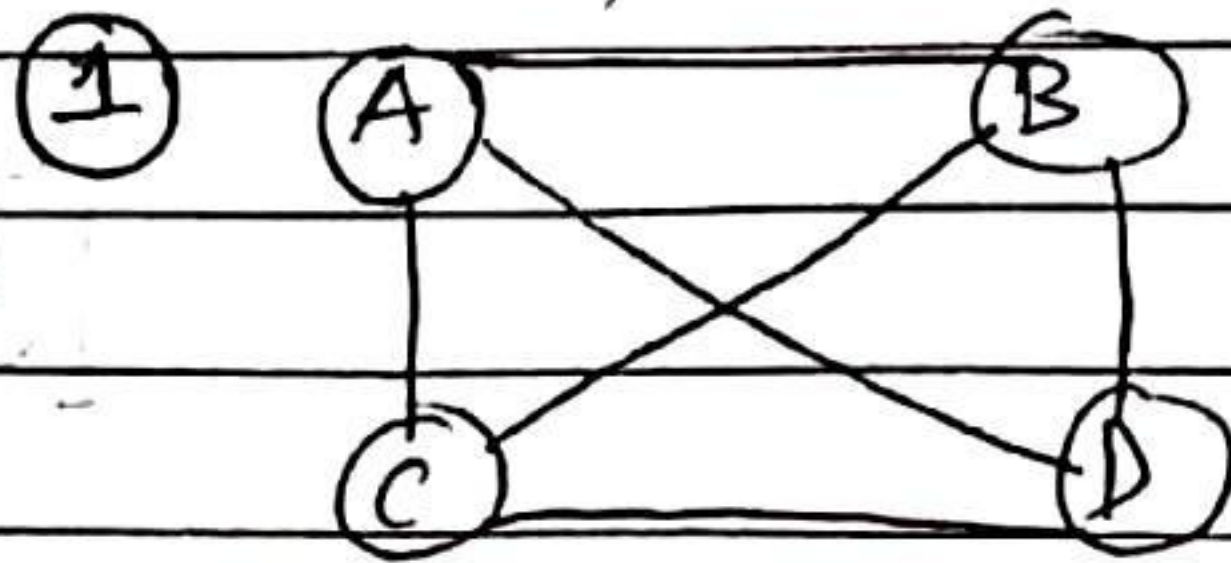


② Unweighted graph —



given

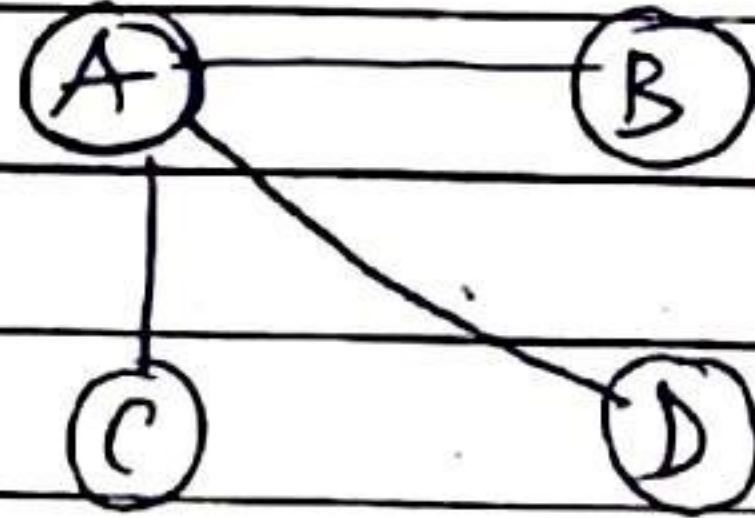
Q. to how many edges required to connect n nodes -



(minimum how many edge required to connect A, B, C, D - 4 nodes)

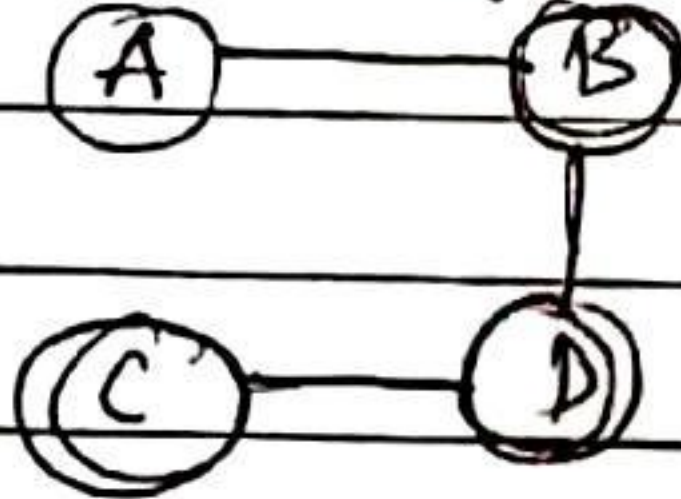
Spanning tree.

→ (3-edge)



→ ~~Span~~ spanning tree.

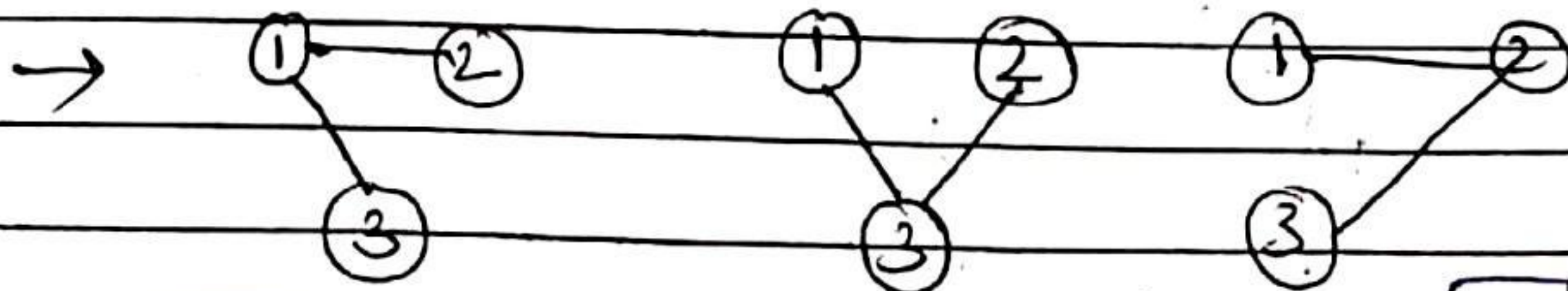
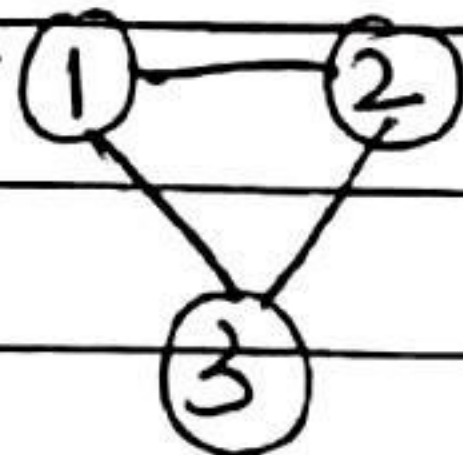
(3-edge)



→ If with To connect ' v ' edges. the min no. of ' E ' edge required $(v-1)$.

→ spanning tree is subgraph. @

Q. ~~How many~~ ^{Max no. of} spanning Tree possible - to given simple leveled ^{complete} graph -



With n vertices leveled vertices n spanning tree possible

$K_4 \rightarrow 4$ node complete graph
 $K_3 \rightarrow 3$ " " " "

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$K_4 \rightarrow 16$ (ST possible)

$K_3 \rightarrow 3$ (ST possible)

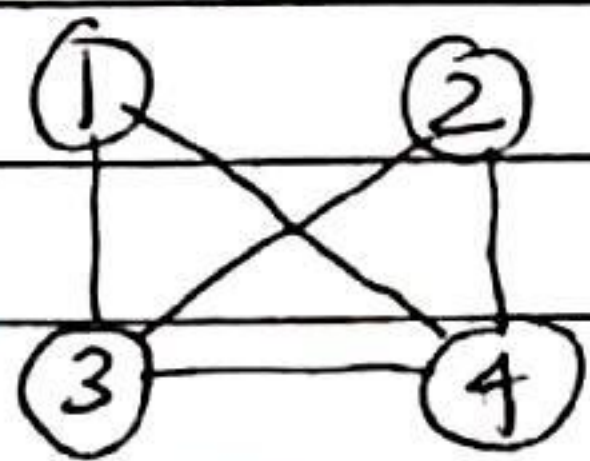
$n \rightarrow$ no. of vertices

$K_n = n^{n-2} \rightarrow$ (for complete graph)

not complete

• Max no. of spanning tree possible to a graph =

ex: 1



(undirected graph)

using

[Kirchoff theorem] =

adjacency matrix for the given graph =

	1	2	3	4
1	0	0	1	1
2	0	0	1	1
3	1	1	0	1
4	1	1	1	0

Kirchoff -
rule -

(i) Take Diagonal 0's $\xrightarrow{\text{replace}}$ Degree of node.

(ii) Non diagonal 1's $\xrightarrow{\text{replace}}$ "-1"

(iii) Non diagonal 0's $\rightarrow$ "0's" :-

Apply this

rule on

adj matrix

	1	2	3	4
1	2	0	-1	-1
2	0	2	-1	-1
3	-1	-1	3	-1
4	-1	-1	-1	3

NO. of spanning Tree (ST) = Cofactor of any element

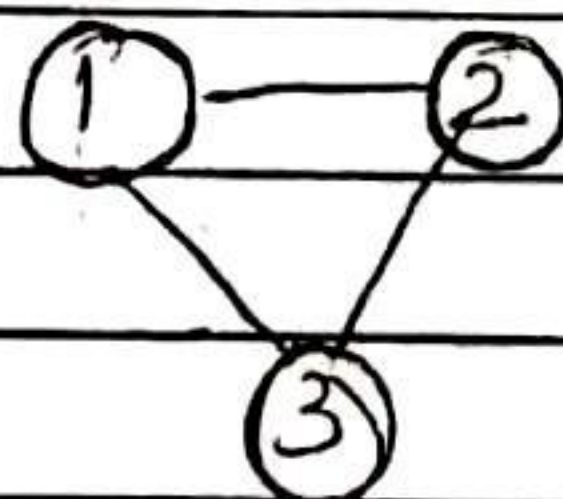
	1	2	3	4
1	2	0	-1	-1
2	0	2	-1	-1
3	-1	-1	3	-1
4	-1	-1	-1	3

Cofactor of 2 is
remove the row and
remove the column

$$\begin{aligned}
 \# \text{Cofactor} &= 2(9-1) - (-1)(-3-1) + (-1)(1+3) \\
 &= 16 + (-4) - 4 \\
 &= 16 - 8 \\
 &= 8
 \end{aligned}$$

So, max no. of spanning trees possible = 8. Present in given graph

Ex: 2 find max ST = ?



Kirchhoff theorem =

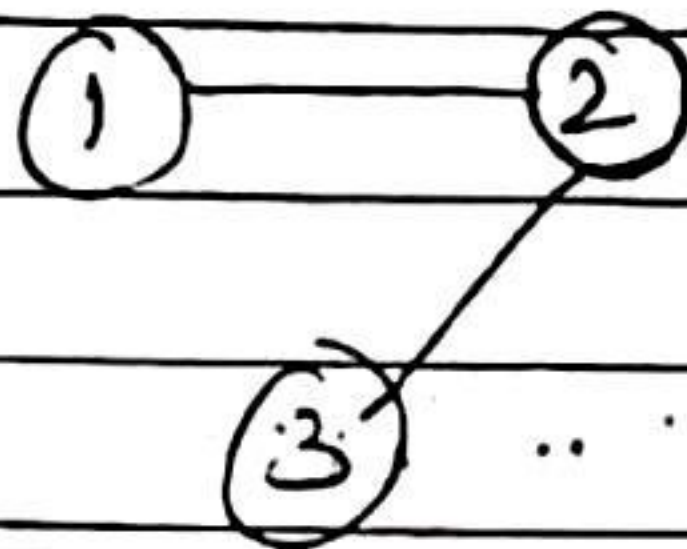
$$\text{adj matrix} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

then, apply 3-rules of Kirchhoff —

$$\begin{aligned}
 &= \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \end{matrix} \\
 &= \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} \\
 &= 4 - 1 = 3
 \end{aligned}$$

So, no. of ST present = 3 ✓

Ex:3 find ~~ST~~? number of ST = ?



Using

Kirchhoff theorem

adj matrix =

	1	2	3
1	0	1	0
2	1	0	1
3	0	1	0

Apply 3-rule =

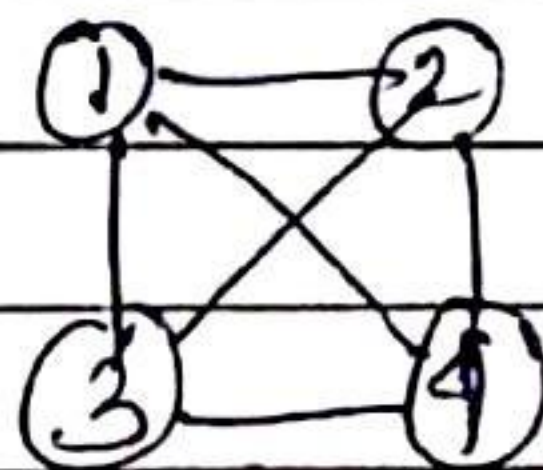
	1	2	3
1	1	-1	0
2	-1	2	-1
3	0	-1	1

$$= \begin{vmatrix} 2 & -1 \\ -1 & 1 \end{vmatrix} = 2 - 1 = 1$$

So, no. of ST present on given graph = 1

Ex:4

no. of ST = ?



adj adjacency matrix =

	1	2	3	4
1	0	1	1	1
2	1	0	1	1
3	1	1	0	1
4	1	1	1	0

then apply Kirchhoff 3-rules =

	1	2	3	4
1	3	-1	-1	-1
2	-1	3	-1	-1
3	-1	-1	3	-1
4	-1	-1	-1	3

$$= \begin{vmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{vmatrix}$$

$$= 3(9 - 1) + (-3 - 1) + (-1)(1 + 3)$$

$$= 24 + (-4) + (-4)$$

$$= 24 - 8$$

$$= 16$$

S.T present in given graph = 16.

OR

$$S.T = n^{n-2}$$

$$= 4^{4-2}$$

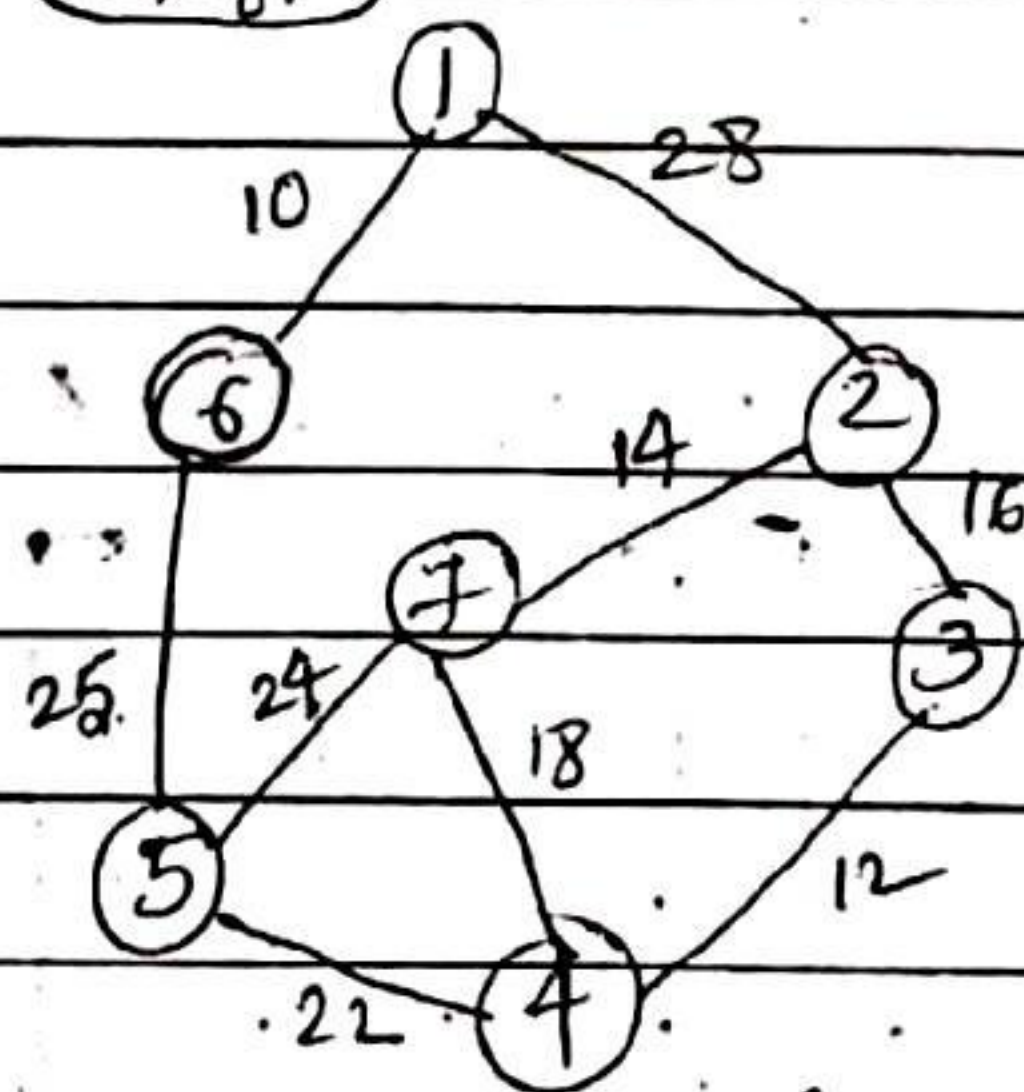
$$= 16$$

→ apply when given graph is Complete graph.

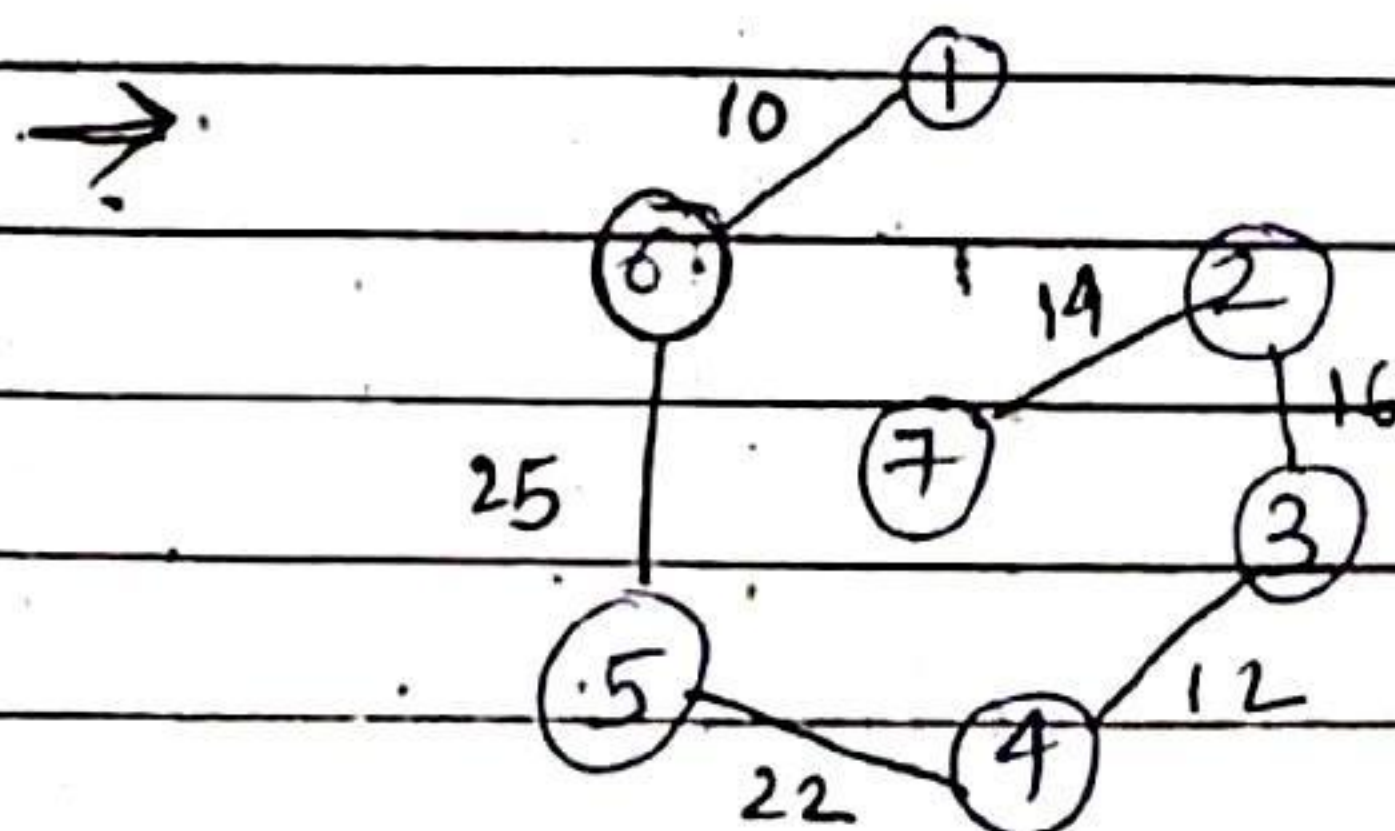
• Minimum Cost Spanning Tree problem =

• Find minimum cost spanning Tree using PRIMS Algorithm.

Ex 81



find minimum then add

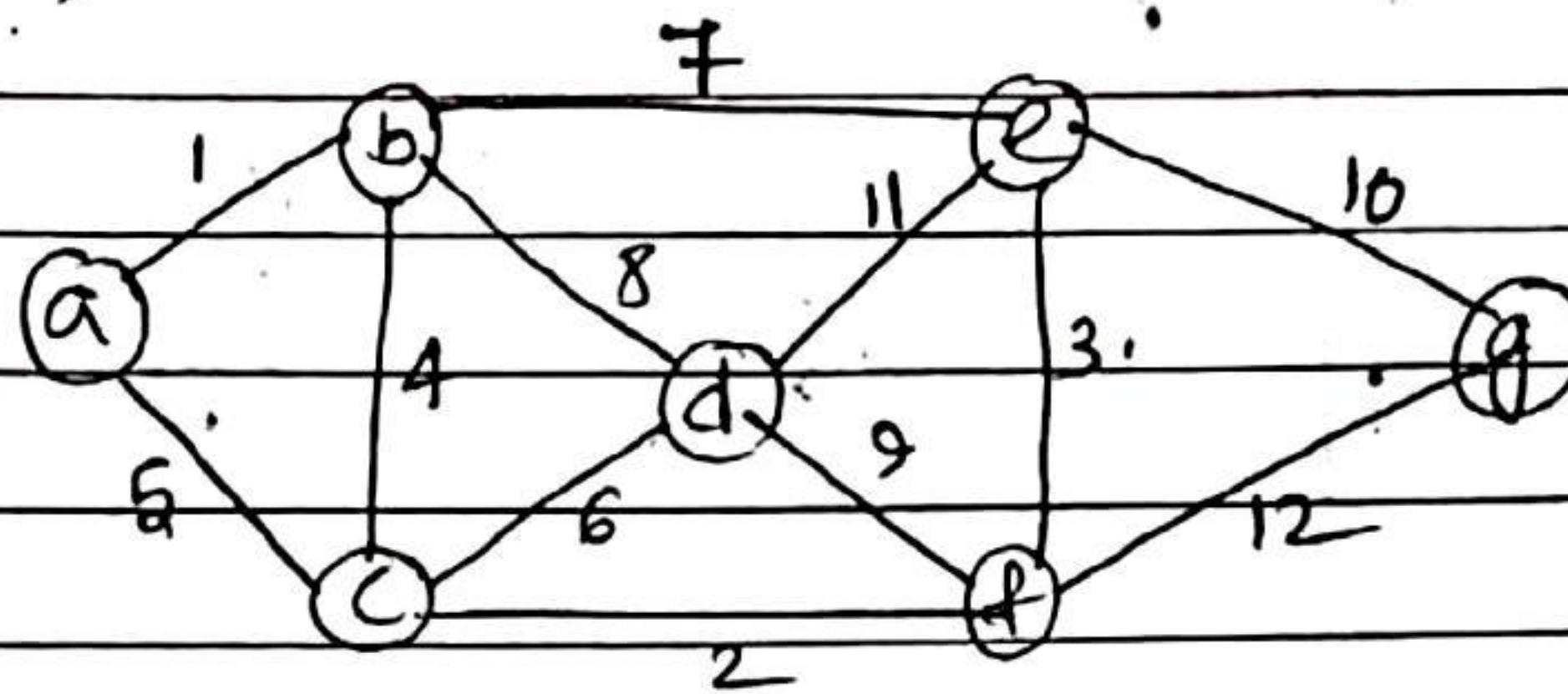


S-1 : 1st Draw all the vertex

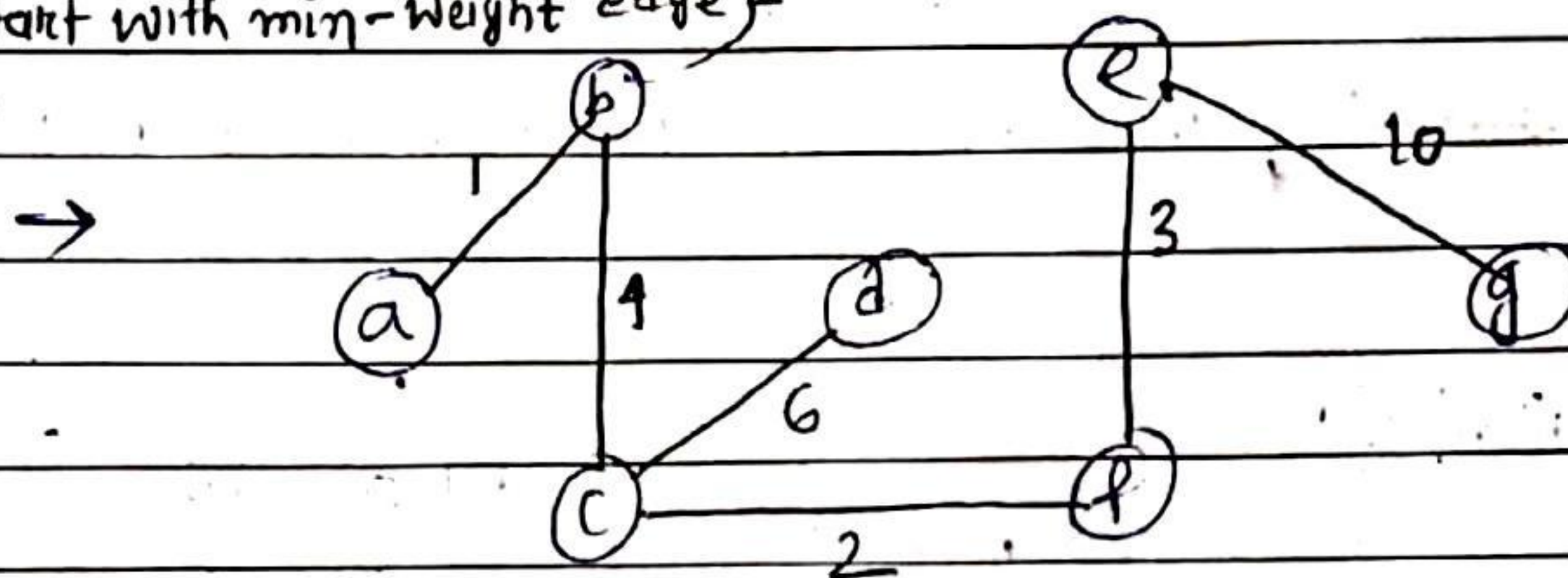
S-2 : Connect by min weight edge.

ex: 2

Given,



(Start with min-weight edge)

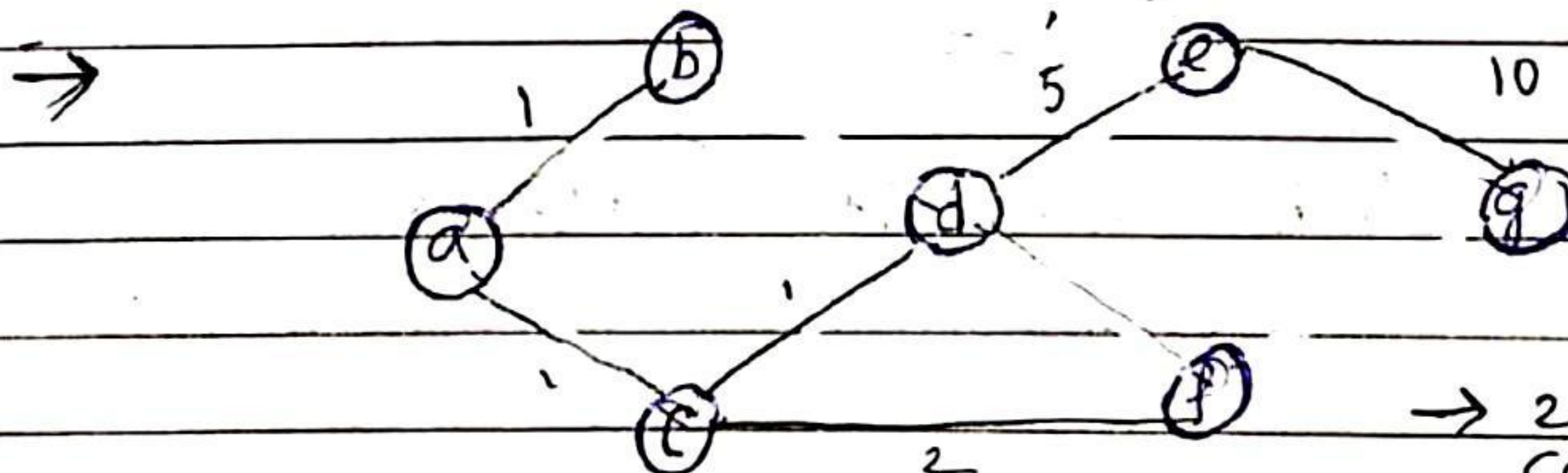
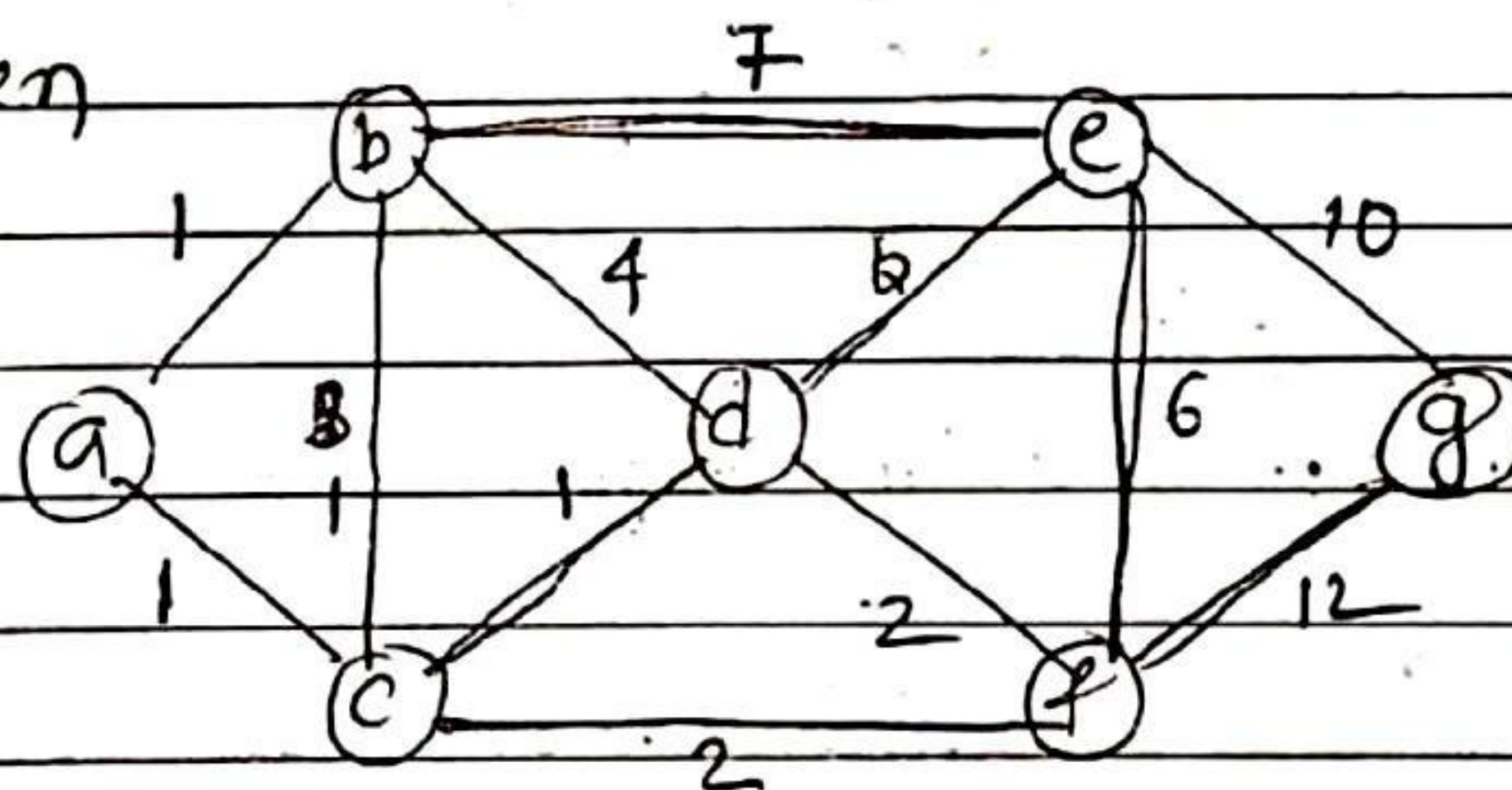


grow step by step one node at a time.

→ here you not will get cycle.

ex-3

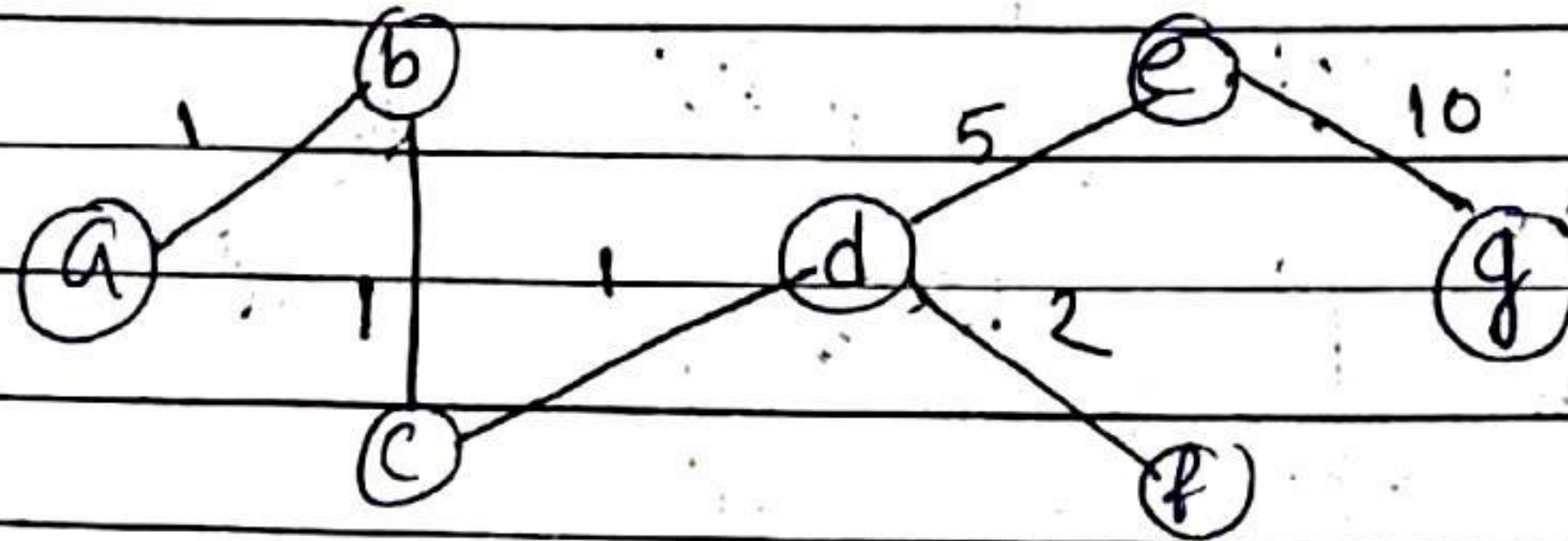
Given



→ 20 is the min cost ST.

→ Whenever we have same weights for more than one edges there is a chance that we might get more than

One spanning tree but any of the cases the final answer is going to be same.



→ 20 min cost ST.

~~Spanning tree are~~ Both the cases cost will be same.

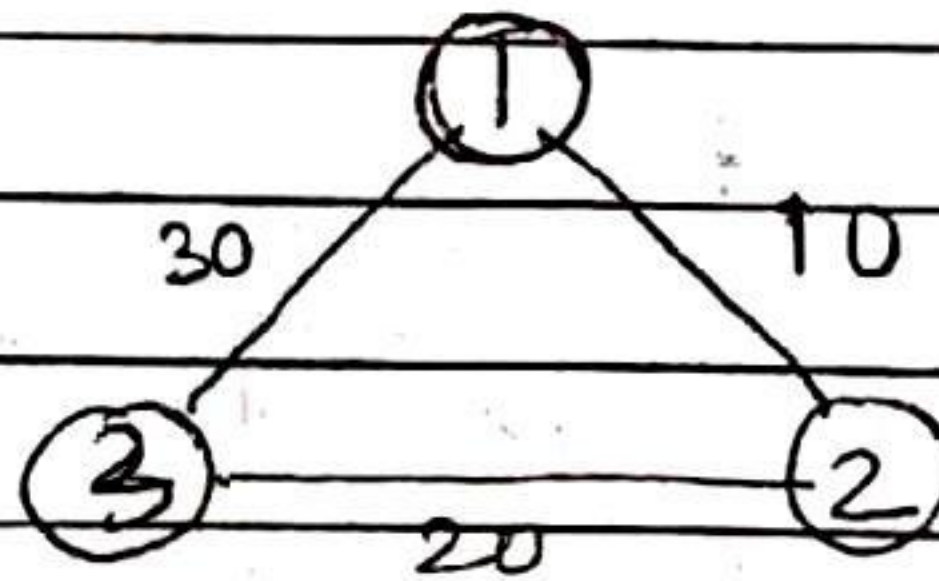
Question - (1)

given cost matrix =

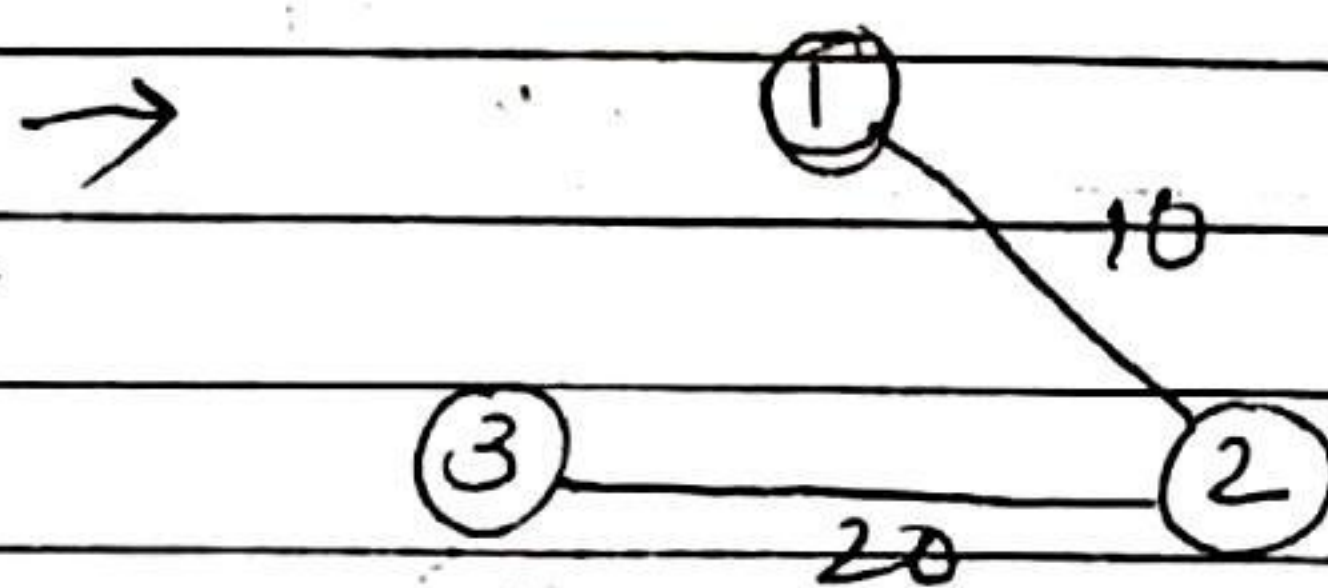
	1	2	3
1	0	10	30
2	10	0	20
3	30	20	0

Find min cost spanning Tree?

→ Draw the graph using cost matrix =



then apply prim's Algorithm =

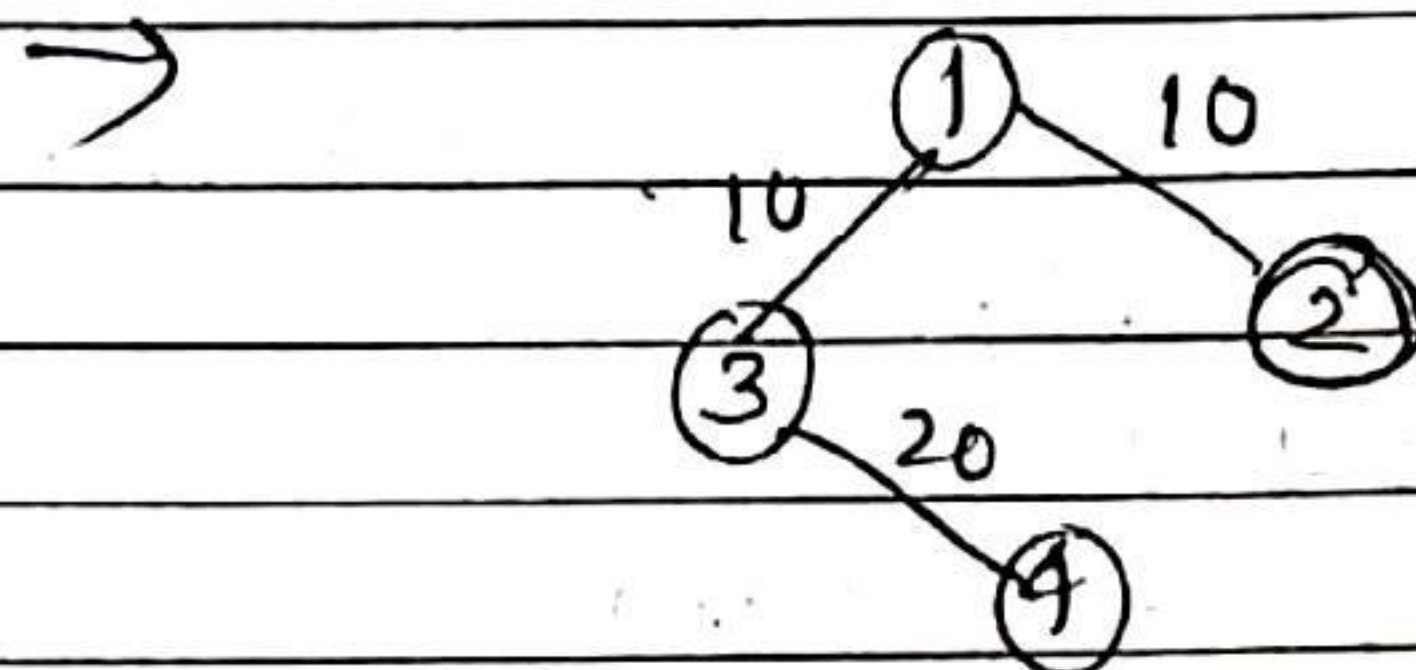


Total min cost = $20 + 10 = 30$

Question - ②

	1	2	3	4
1	0	10	10	50
2	10	0	40	30
3	10	40	0	20
4	50	30	20	0

Find min cost ST?



Total cost = 40

Q-2410

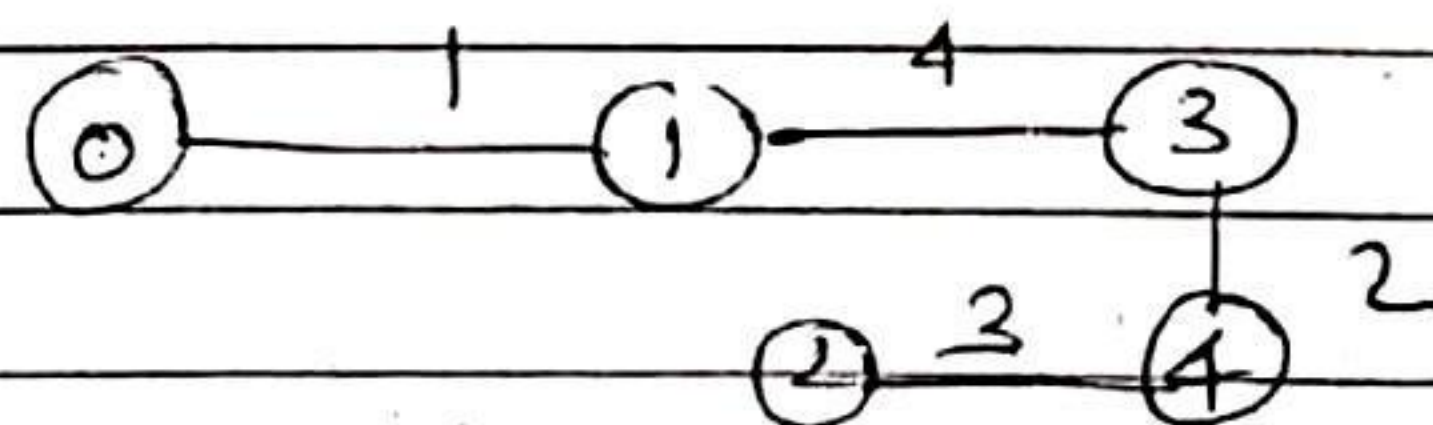
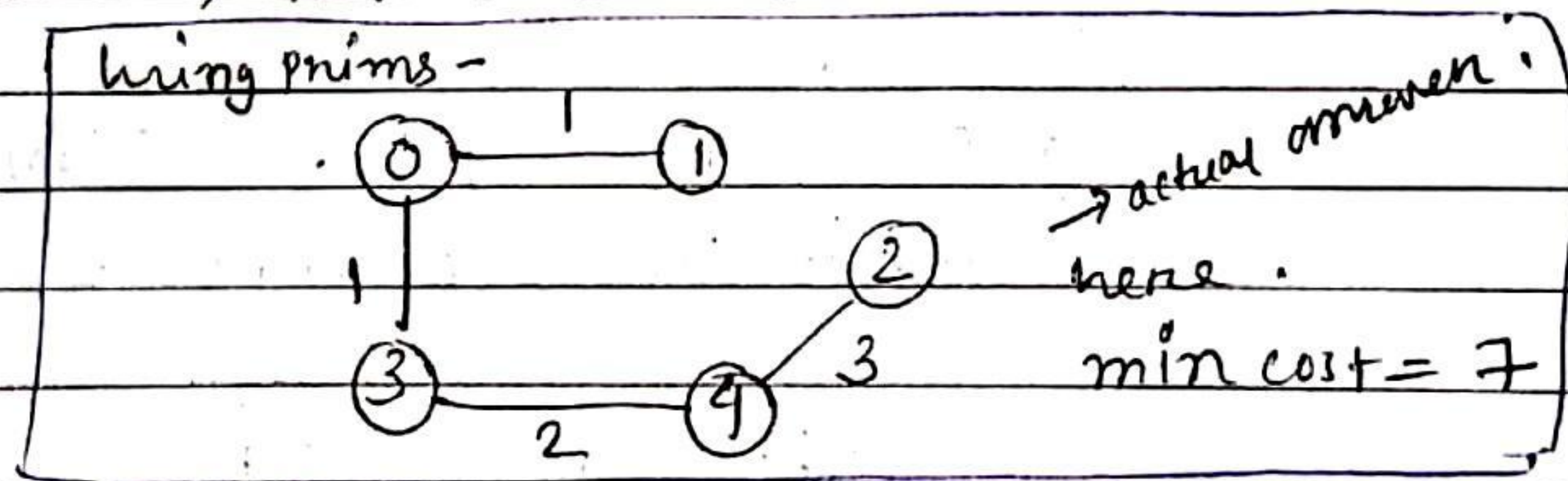
Question - ③

	0	1	2	3	4	
0	0	①	8	①	4	✓✓
1	①	0	12	4	9	✓
2	8	12	0	7	3	✓✓
3	1	4	7	0	2	✓✓
4	4	9	③	②	0	✓✓

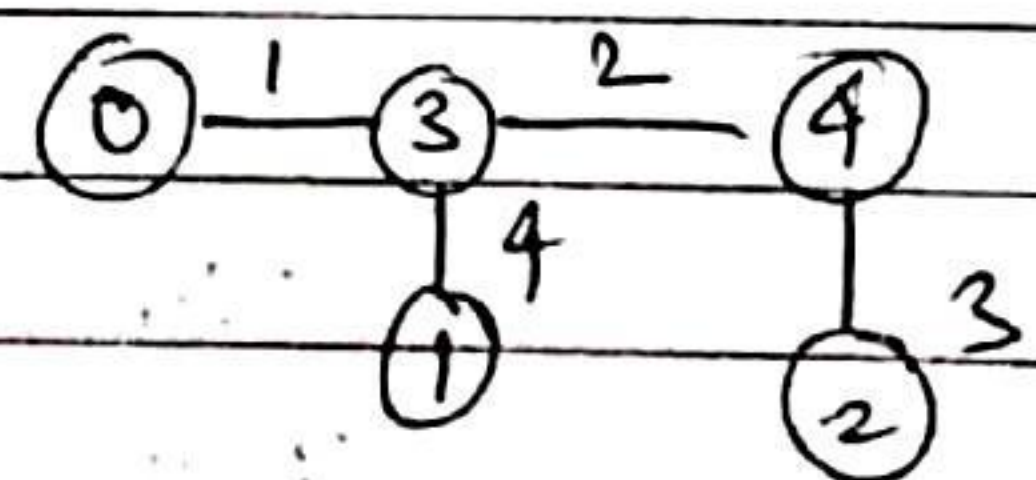
Construct min cost ST

such a way that '0' is a leaf.

Using Prim's -



∴ here cost = 10



here cost = 10

more than one spanning tree but cost is same 10.
 according to question 10 is cost.

Implementation

- PRIMS Algorithm (without min heap) $n \rightarrow$ no of vertices

Implementation with

Algorithm Prim($E, cost, n, t$)// E is the set of edges. $cost$ is $(n \times n)$ adjacency matrix// MST is computed and stored in array $t[1:n-1, 1:2]$.

{

1. let (k, l) be an edge of min cost in E ; $\rightarrow O(E)$ 2. $mincost = cost[k, l]$;3. $t[1, 1] = k$; $t[1, 2] = l$;4. for $(i = 1 \text{ to } n)$ $\rightarrow O(n)$ 5. if $(cost[i, 1] < cost[i, k])$ then $near[i] = 1$;6. else $near[i] = k$;7. $near[k] = near[l] = 0$;8. for $(i = 2 \text{ to } n-1)$ $\rightarrow (n-2)$

{

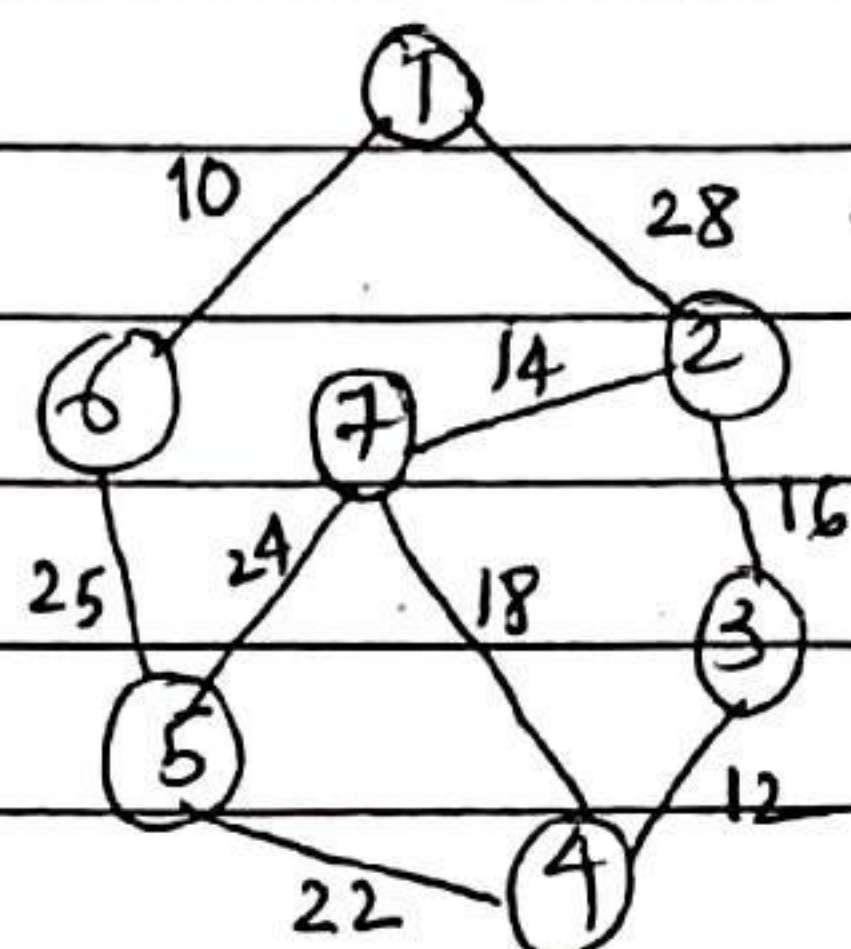
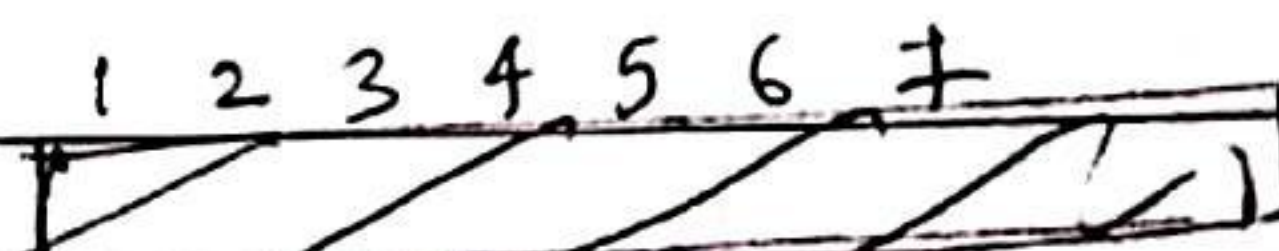
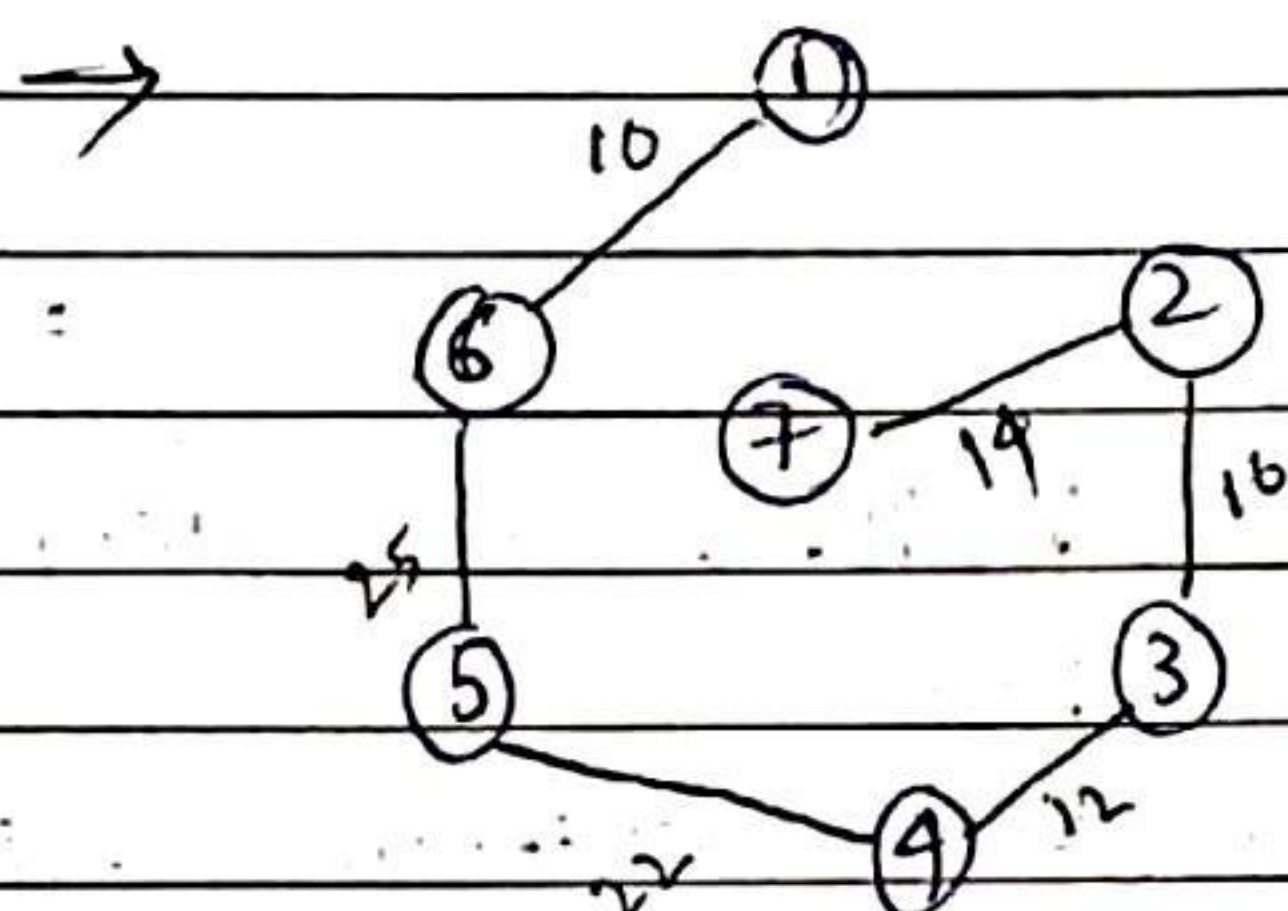
9. let j be an index such that $near[j] \neq 0$ and $cost[j, near[j]]$ is minimum; $\rightarrow O(n)$ 10. $t[i, 1] = j$; $t[i, 2] = near[j]$;11. $mincost = mincost + cost[j, near[j]]$;12. $near[j] = 0$;13. for $k = 1 \text{ to } n$ do $\rightarrow O(n)$ 15. if $(near[k] \neq 0 \text{ and } (cost[k, near[k]] > cost[k, j]))$ then $near[k] = j$;

}

}

Prims algorithm

Time complexity = $O(n^2)$.

ex^o $n \rightarrow$ verticesnear
arraydata structure
should hold all edges
in the spanning
tree.

1	(1,6)
2	(5,6)
3	(4,5)
4	(3,4)
5	(2,5)
6	(7,4)

$$\text{min cost} = 10 + 25 + 22 + 12 + 16 + 14$$

$$= 99$$

closed to

near array

1	2	3	4	5	6	7
1	28	6	6	6	10	6
0	28	∞	∞	∞	(10)	∞

1	2	3	4	5	6	7
1	1	6	6	6	6	6
0	28	∞	∞	(25)	0	∞

1	2	3	4	5	6	7
1	1	6	5	5	6	5
0	28	∞	(22)	0	0	24

1	2	3	4	5	6	7
1	2	3	4	5	6	2
0	0	0	0	0	0	14

1	2	3	4	5	6	7
1	2	3	4	5	6	2
0	0	0	0	0	0	14

1	2	3	4	5	6	7
1	1	4	4	5	6	4
0	28	(12)	0	0	0	18

1	2	3	4	5	6	7
1	3	3	4	5	6	4
0	(16)	0	0	0	0	18

(Add vertices)

- PRIMS algorithm (implement with min heap)

MST-PRIM (G, cost, ∞) // using min heap.

$\{$
 1. for each vertex $u \in G \text{ vertices}$ } — $O(V)$
 $\{$
 2. $u.\text{key} = \infty$
 3. $u.\pi = \text{NIL}$
 $\}$

$G \rightarrow \text{graph}$
 $\text{cost} \rightarrow \text{cost matrix}$
 $\pi \rightarrow \text{root}$

4. $\pi.\text{key} = 0$

$\emptyset \rightarrow \text{min heap}$

5. $\emptyset = G \text{ vertices}$ — Build heap — $O(V)$

6. While ($\emptyset \neq \emptyset$) — $O(V)$

7. $\{ u = \text{Extract-MIN}(\emptyset) \}$ — extracting min — $O(\log V)$

8. for each vertex ' v ' adjacent to ' u '. $O(\log V)$
 $\{$ — $O(V)$

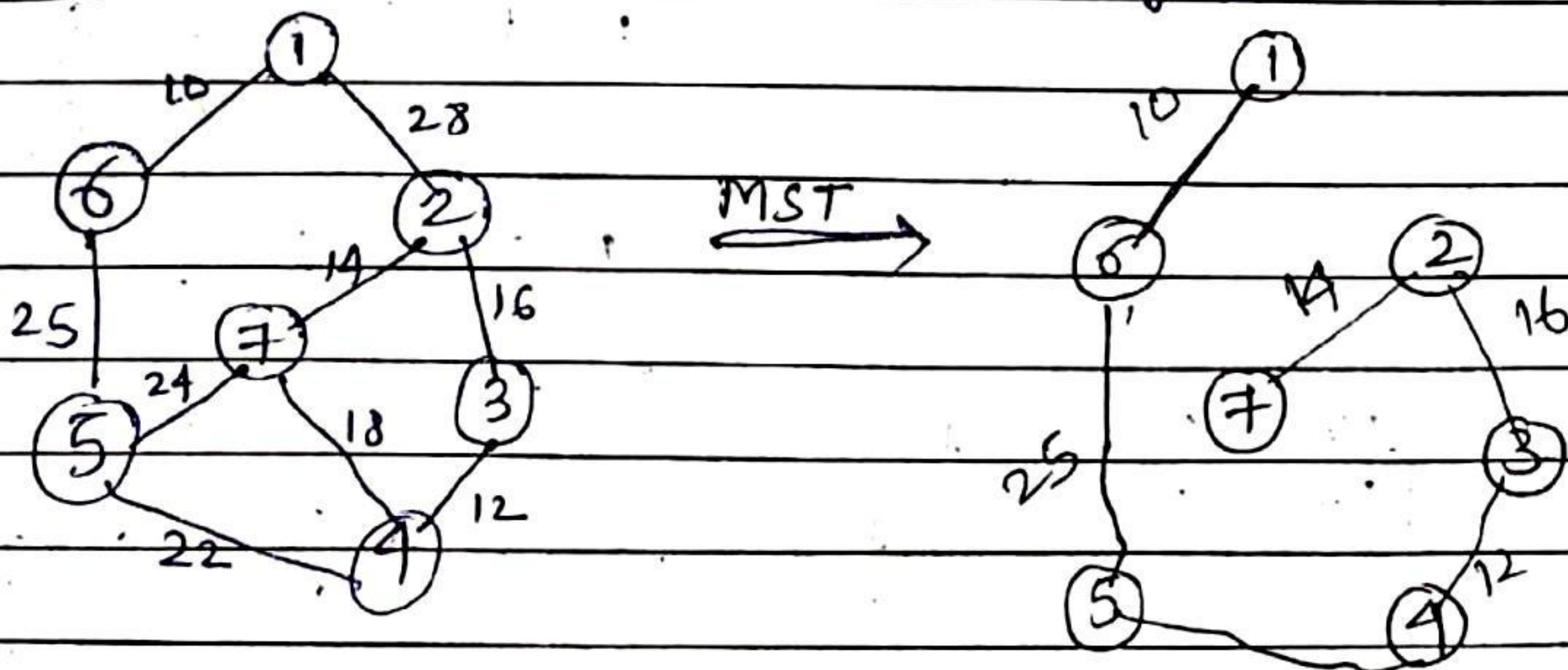
9. if $v \in \emptyset$ and $\text{cost}(u, v) < v.\text{key}$.

10. $v.\text{parent} = u$

11. $v.\text{key} = w(u, v)$ — decrease key — $O(\log V)$
 $\}$

$\}$ ** Time complexity = $O(V \log V + E \log V + E)$
 $= O(E \log V)$.

ex:



Extract min = $O(V \log V)$

Decrease-key = $O(V \cdot V \log V) = O(V^2 \log V) = O(E \log V)$

Build heap = $O(E)$

extract

	1	2	3	4	5	6	7
key	∞	∞	∞	∞	∞	∞	∞
π	N	NIL	N	N	N	N	N

Ex

key	2	3	4	5	6	7
key	28	∞	∞	∞	10	∞
π	1	N	N	N	1	N

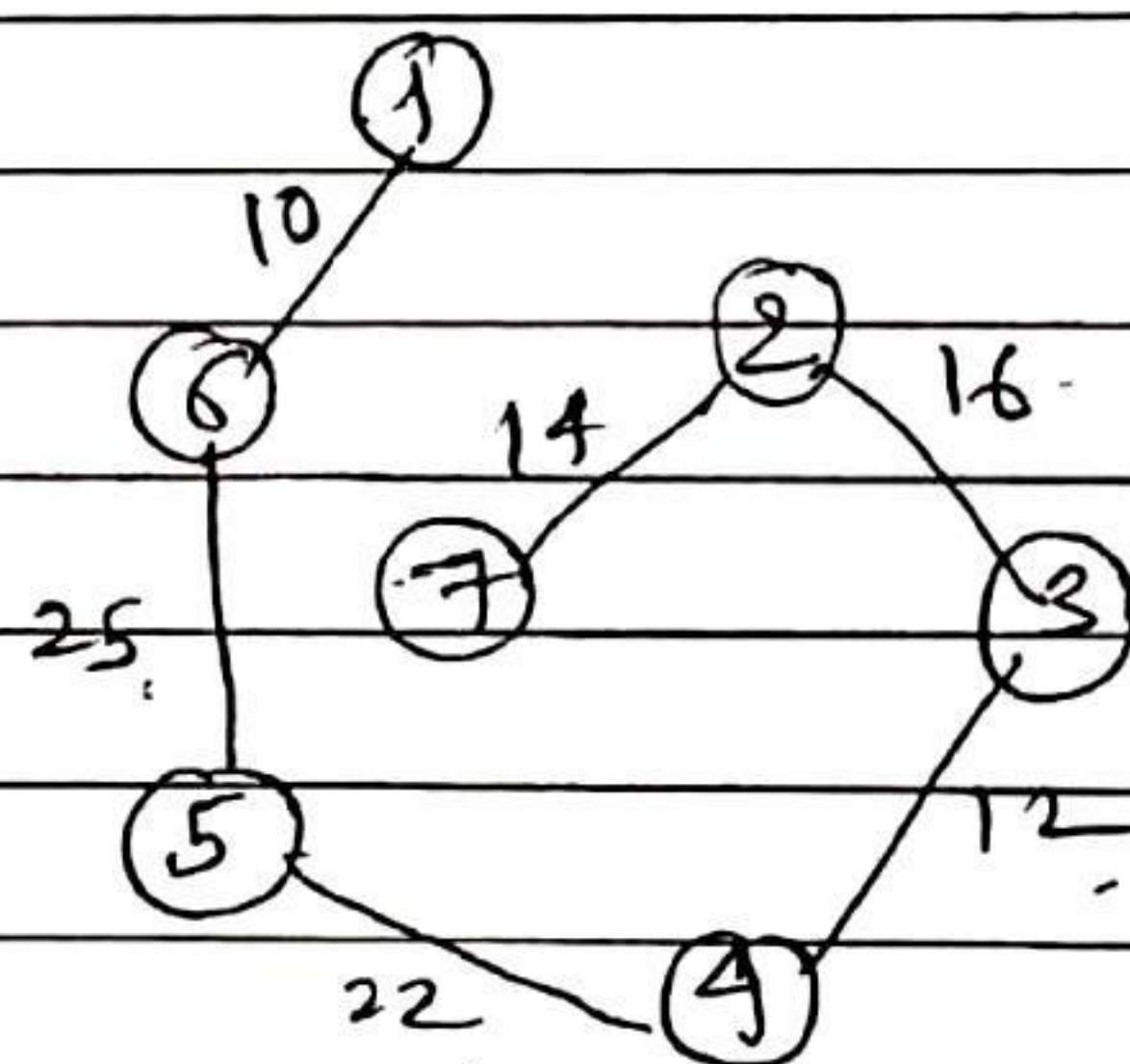
key	2	3	4	5	7
key	28	∞	∞	25	∞
π	1	N	N	6	N

key	2	3	4	7
key	28	∞	22	24
π	1	N	5	5

key	2	3	7
key	28	12	18
π	1	4	4

key	2	7
key	16	18
π	3	4

key	7
key	14
π	2



MST = 99
(min cost spanning tree)

	out		
	With heap	with heap	fibonacci heap
Time Complexity:	$O(m^2)$	$O(E \log v)$	$O(v \log v + E)$
(Prims algo)	$O(v^2)$		

Comparison — (prime Algo)

• with^{out} heap — $O(V^2)$

• With out heap — $O(E \log V)$

In case of Dense graph: (complete graph) (means, no. of edge E very high)
 $E = O(V^2)$

✓ without heap — $O(V^2)$

with heap — $O(V^2 \log V)$

In case of sparse graph: (no. of E is less)
 $E = O(V)$

with out heap = $O(V^2)$

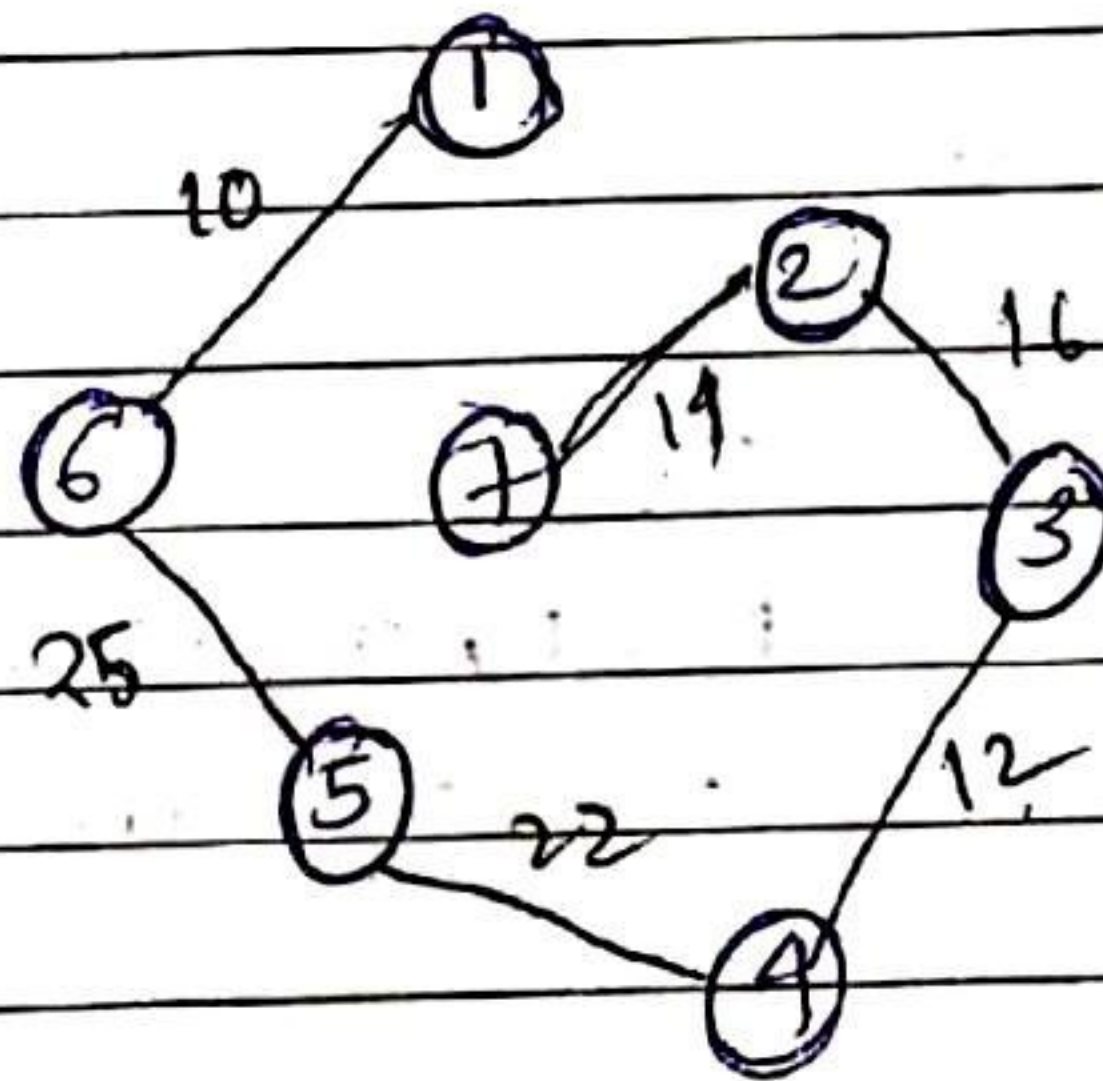
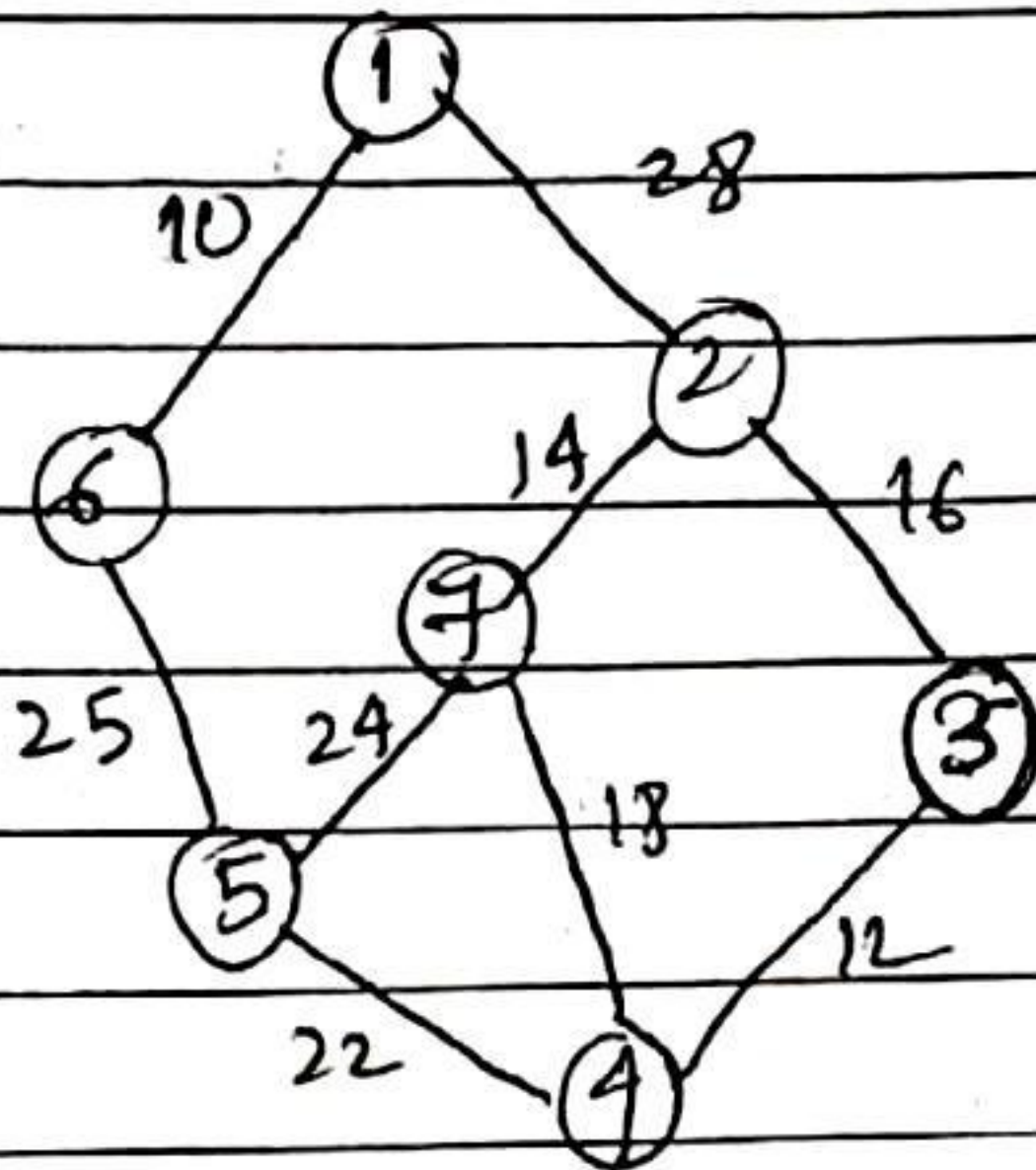
✓ with heap algo = $O(V \log V)$

(add edges)

• Kruskal's algorithm =

ex: 1

{ In Prim we get trees.
but in Kruskal we might get -
forest.



(Connect by : smallest weight)

10, 12, 14, 16, ~~18~~, 22,
~~24~~, 25

MST = 100

→ Cost of spanning tree same both the cases
(prim's and kruskal's)

gate-
2000Question - (1)

Let 'G' be an undirected connected graph with distinct edge weights. Let max_e be the edge with max weight and min_e be the edge with minimum weight. Which of the following is false?

a) Every minimum spanning tree of 'G' must contain min_e edge.

b) If max_e is in a minimum spanning tree, then

its removal must disconnect G .

~~False~~ No minimum spanning tree contains max edge.

4) G has a unique minimum spanning tree.

gates-
2007

Question - 2

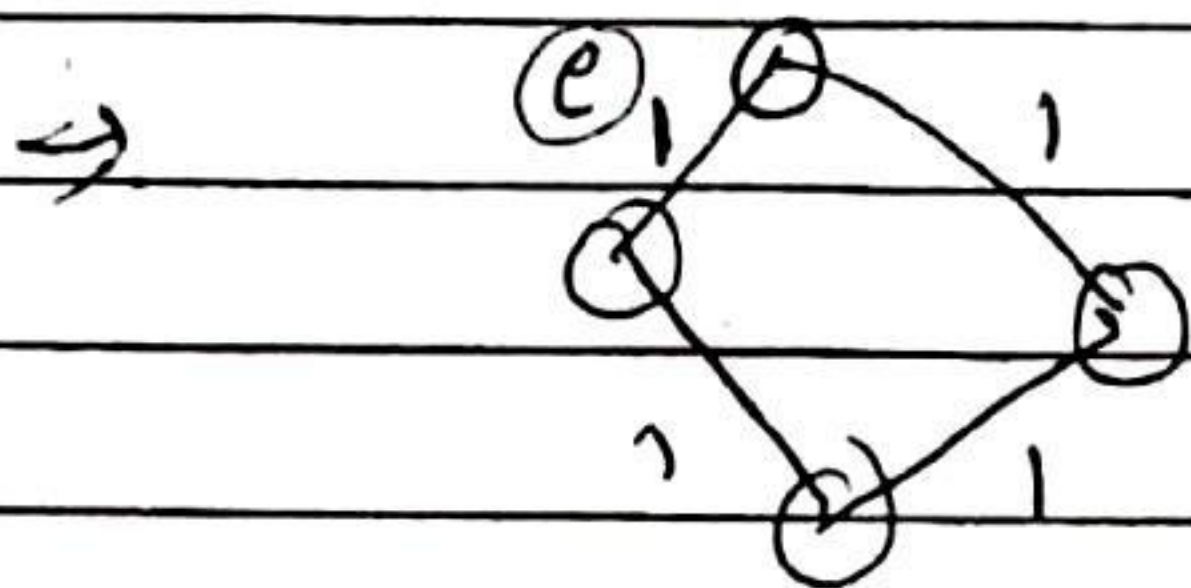
Let's w be the minimum weight among all weights in an undirected connected graph. Let e be a specific edge of weight w . Which of the following is false.

a) There is a minimum spanning tree containing e .

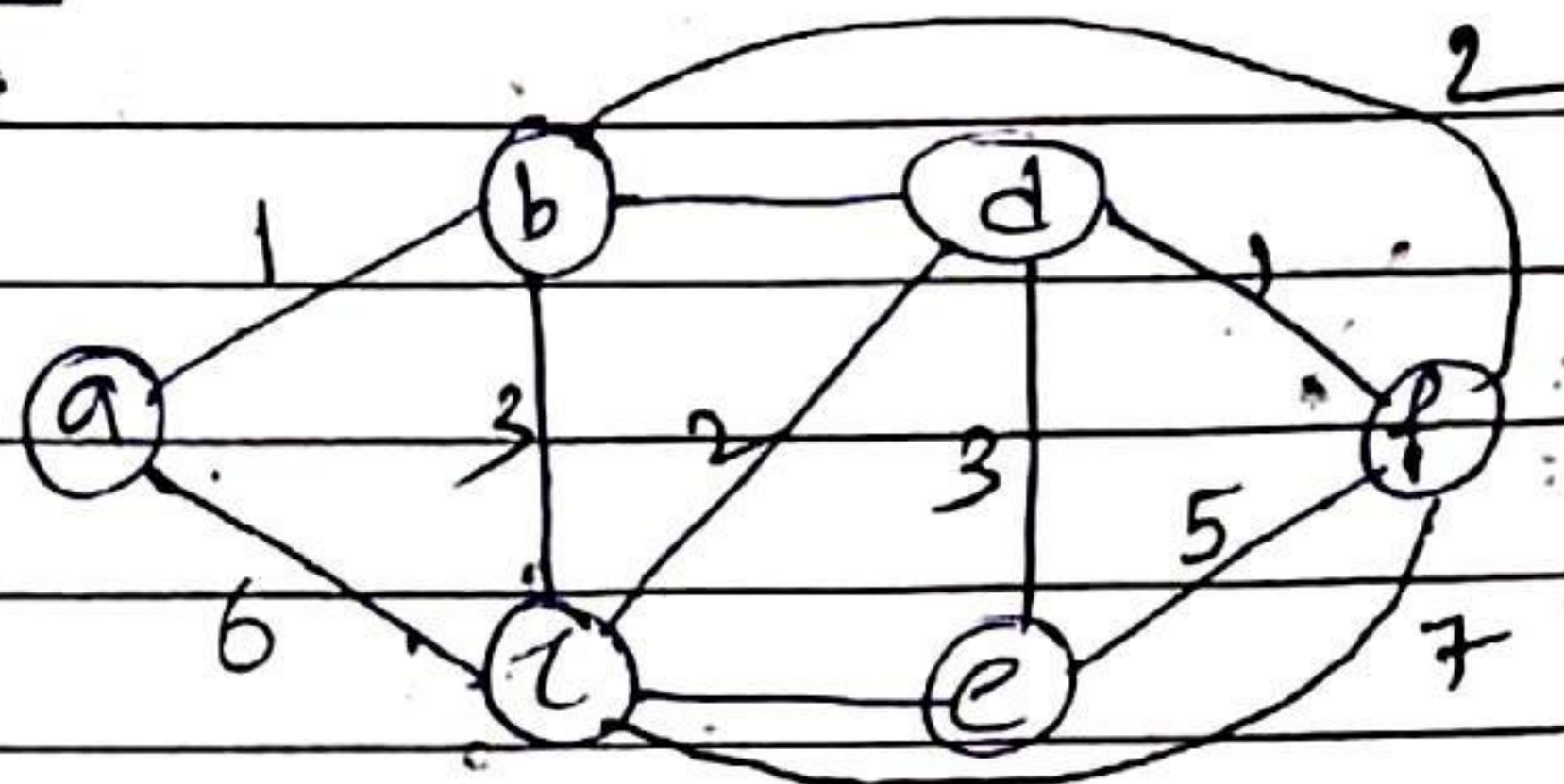
~~False~~ If e is not in MST, T , then in the cycle formed by adding e to T , all edges have the same weight.

c) Every minimum spanning tree has an edge of weight w .

~~False~~ e is present in every minimum spanning tree.



9-2006

Question - 3

Which of the following can't be the sequence of edges added, in that order to minimum spanning tree using Kruskal's algorithm.

a) $\overset{1}{(a-b)}$ $\overset{1}{(d-f)}$ $\overset{2}{(b-f)}$ $\overset{2}{(d-c)}$ $\overset{3}{(d-e)}$

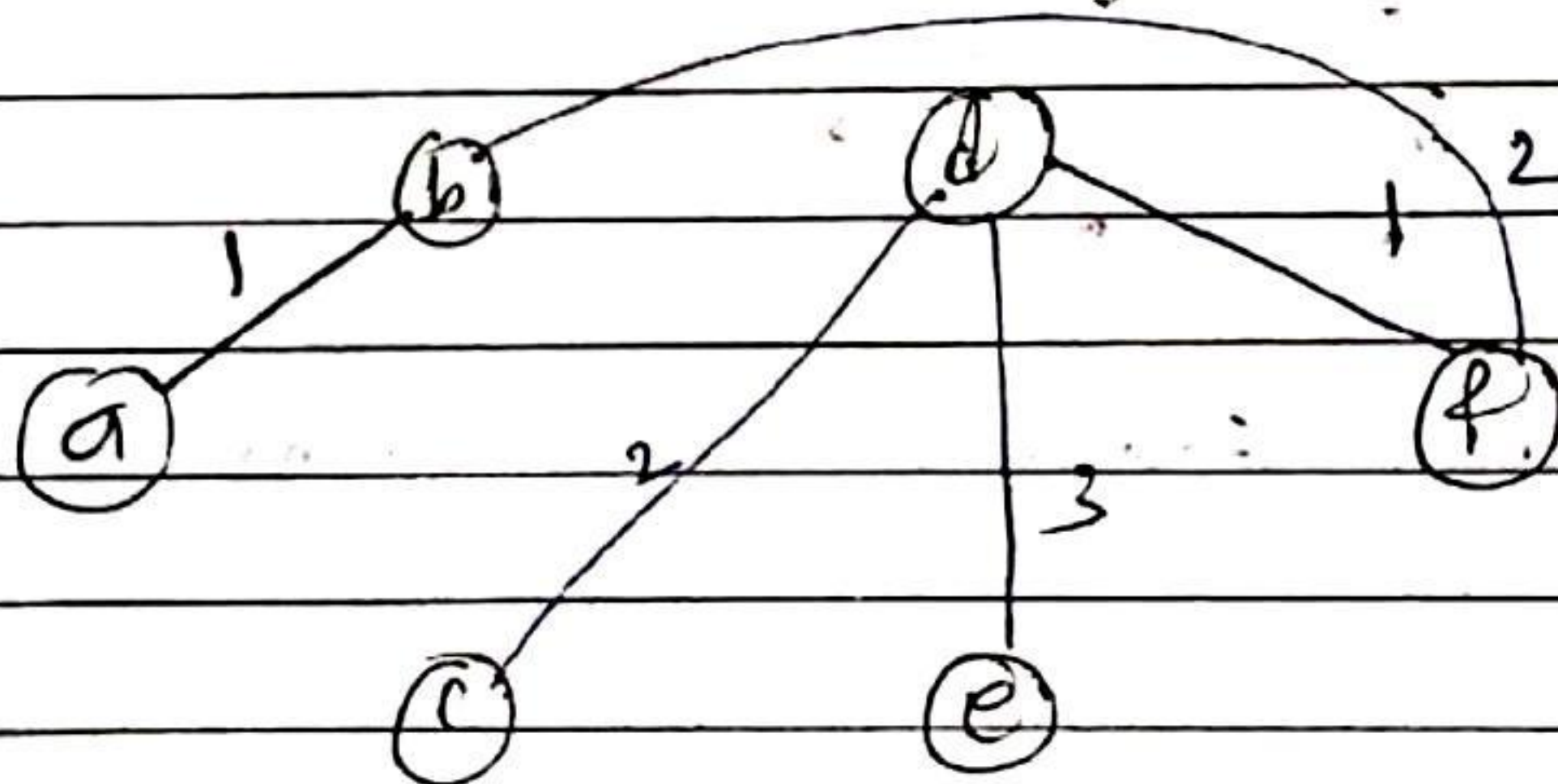
b) $\overset{1}{(a-b)}$ $\overset{1}{(d-f)}$ $\overset{2}{(d-c)}$ $\overset{2}{(b-f)}$ $\overset{3}{(d-e)}$

c) $\overset{1}{(d-f)}$ $\overset{1}{(a-b)}$ $\overset{2}{(d-c)}$ $\overset{2}{(b-f)}$ $\overset{3}{(d-e)}$

(lowest to highest)

d) $\overset{1}{(d-f)}$ $\overset{1}{(a-b)}$ $\overset{2}{(b-f)}$ $\overset{3}{(d-e)}$ $\overset{2}{(d-c)}$

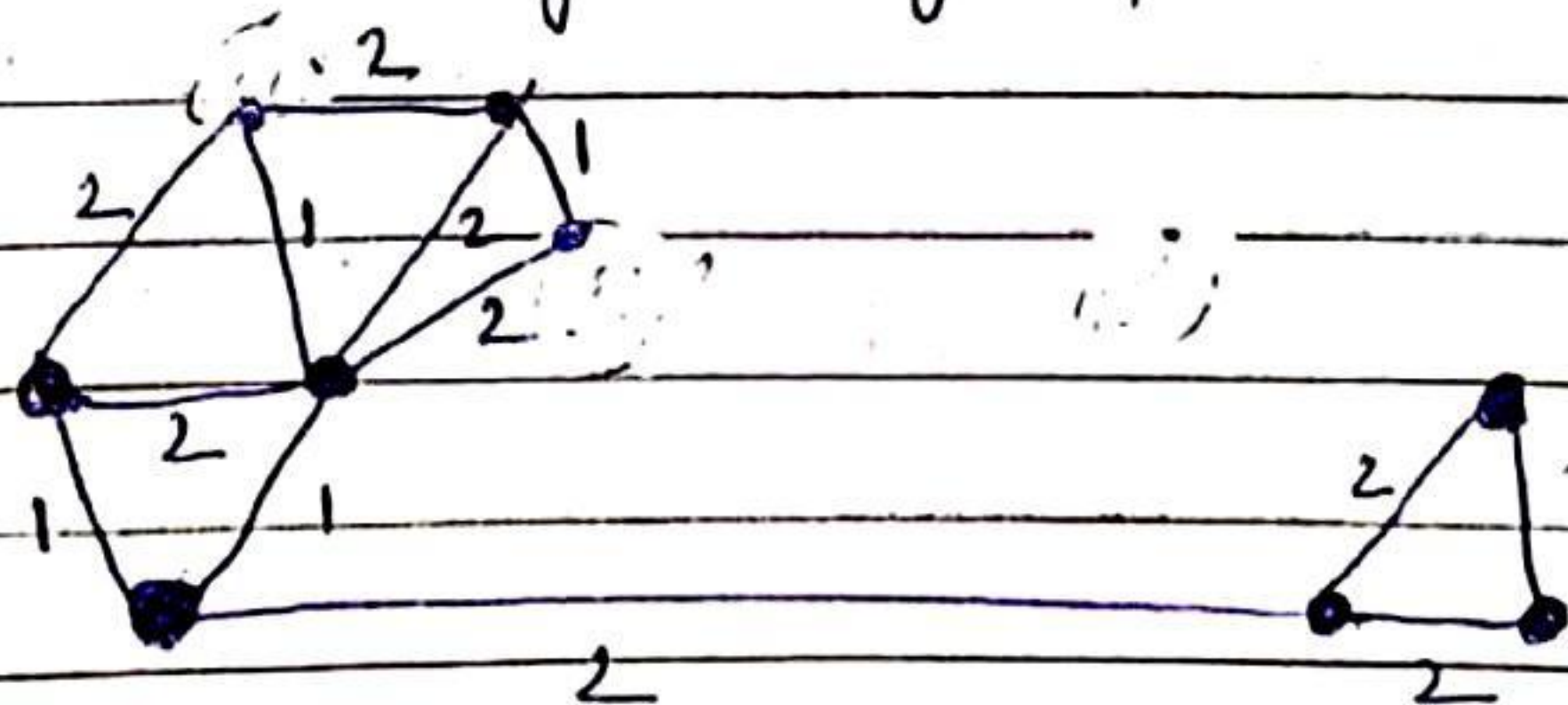
Wrong



9-2014

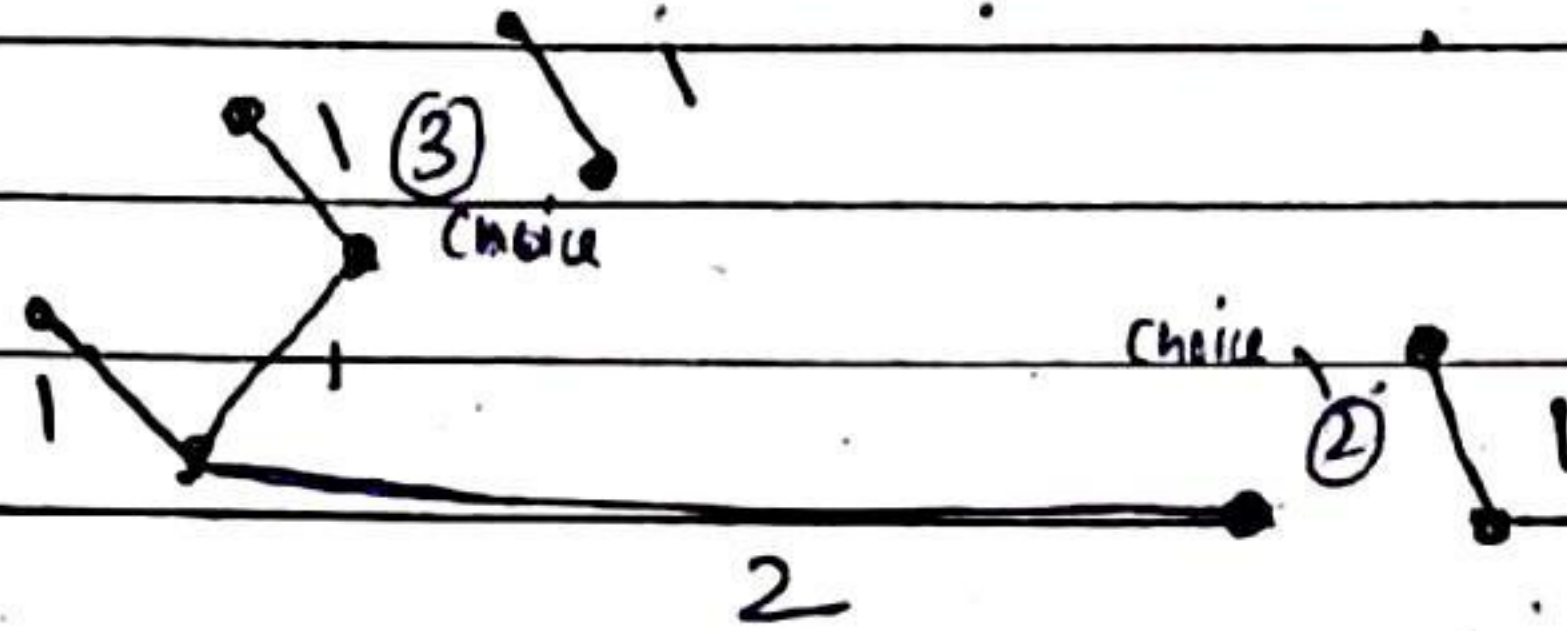
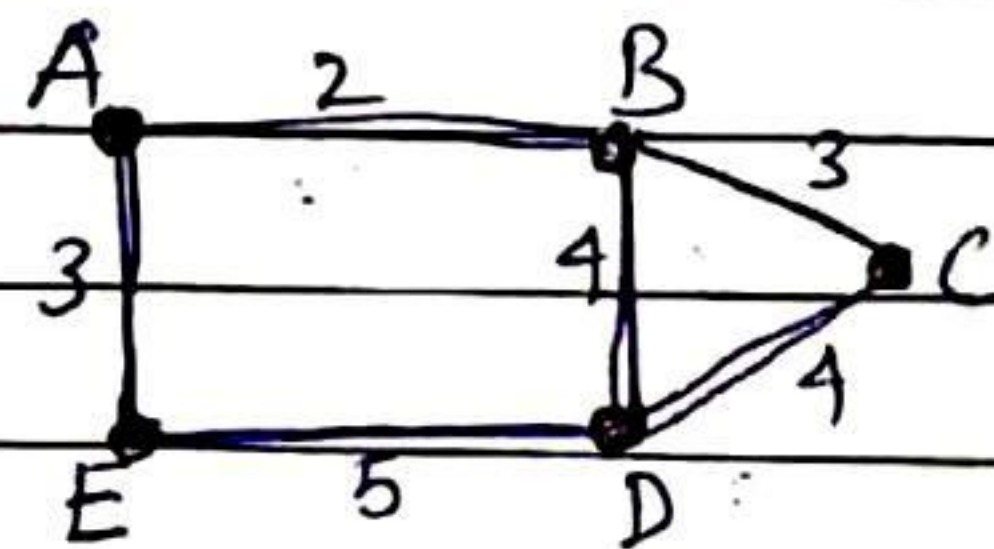
Question - 4

The no. of distinct minimum spanning trees for the weighted graph below is -

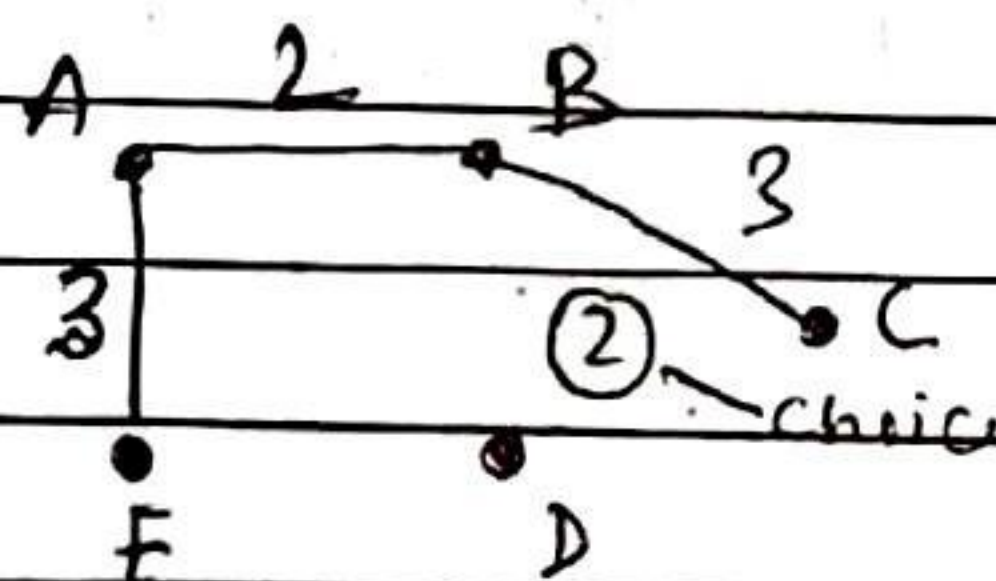


→

first go with less weighted edge.

Total spanning tree possible = $2 \times 3 = 6$ Question - (5)

How many spanning trees possible?



Total minimum spanning tree possible = (2)

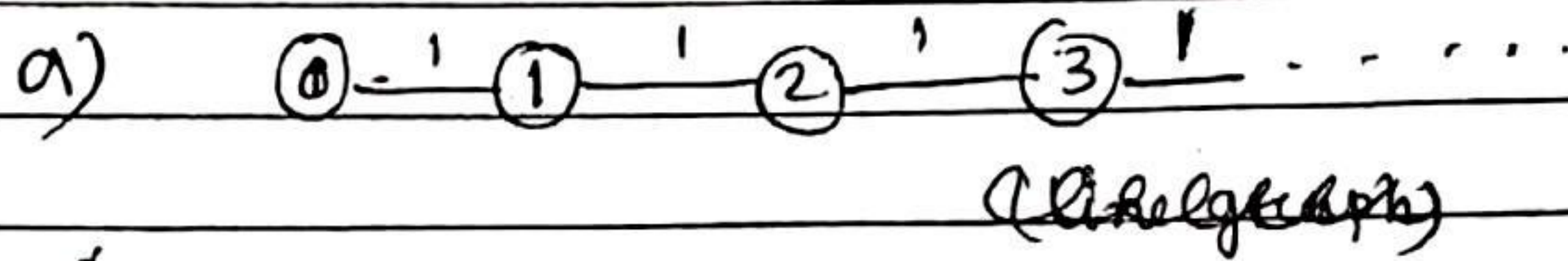
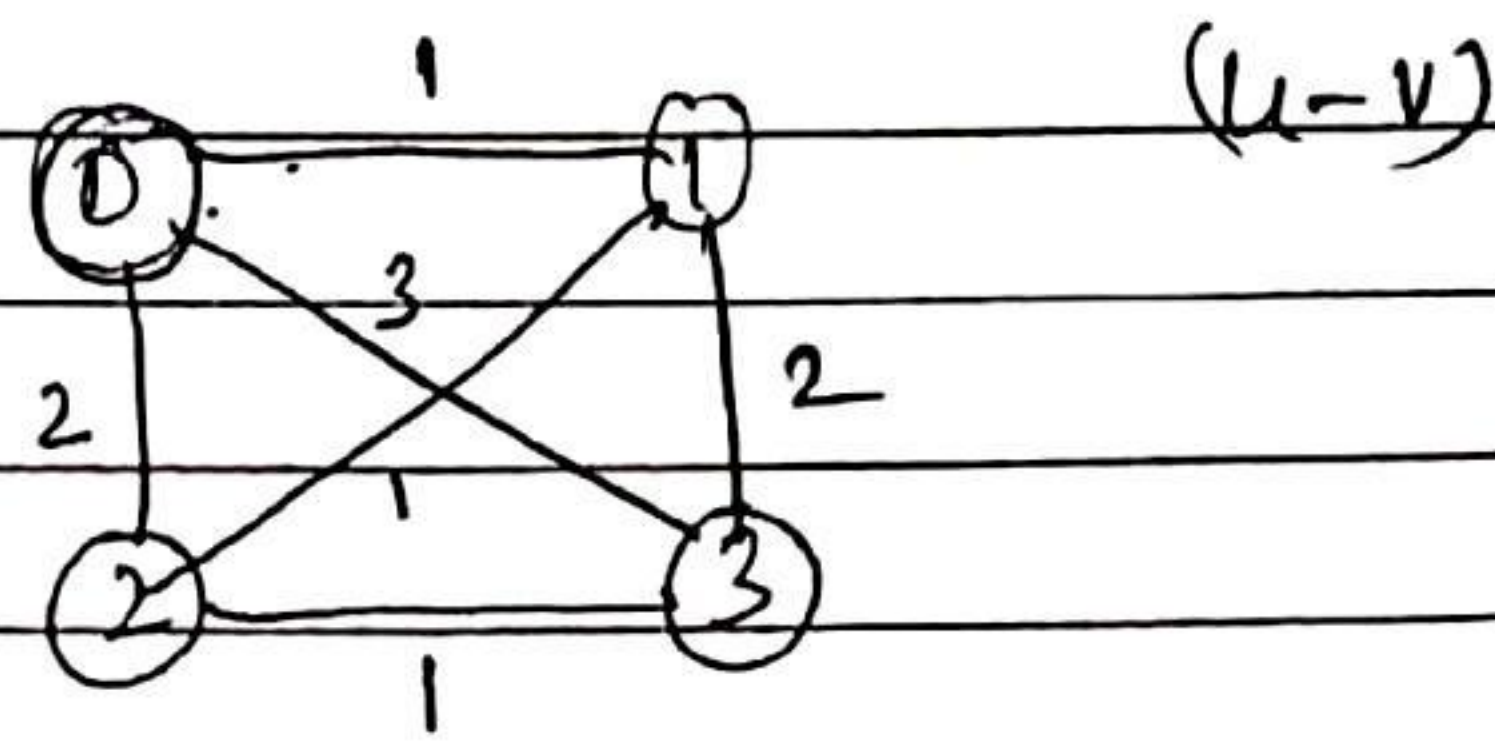
Question - (6)

A complete, undirected, weighted graph 'G' is given on the vertex $\{0, 1, \dots, n-1\}$ for any fixed 'n'. Draw the minimum spanning tree of G if -

a) The weight of the edge (u, v) is $|u - v|$.

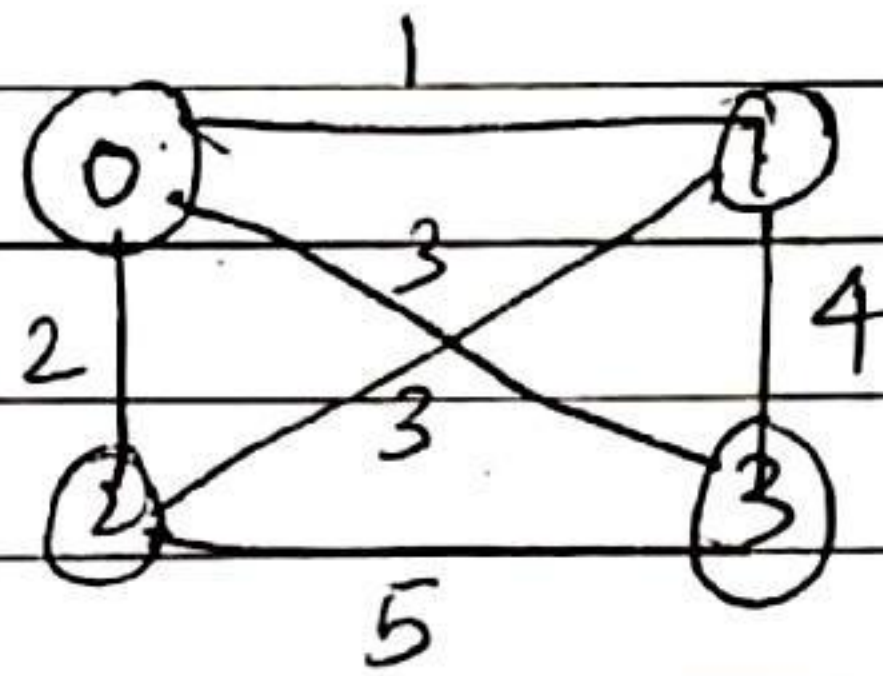
b) The weight of the edge (u, v) is $(u + v)$.

→ $n=4$



~~b)~~

$m=4$



b)

